

THE THAMES
AND ITS DOCKS

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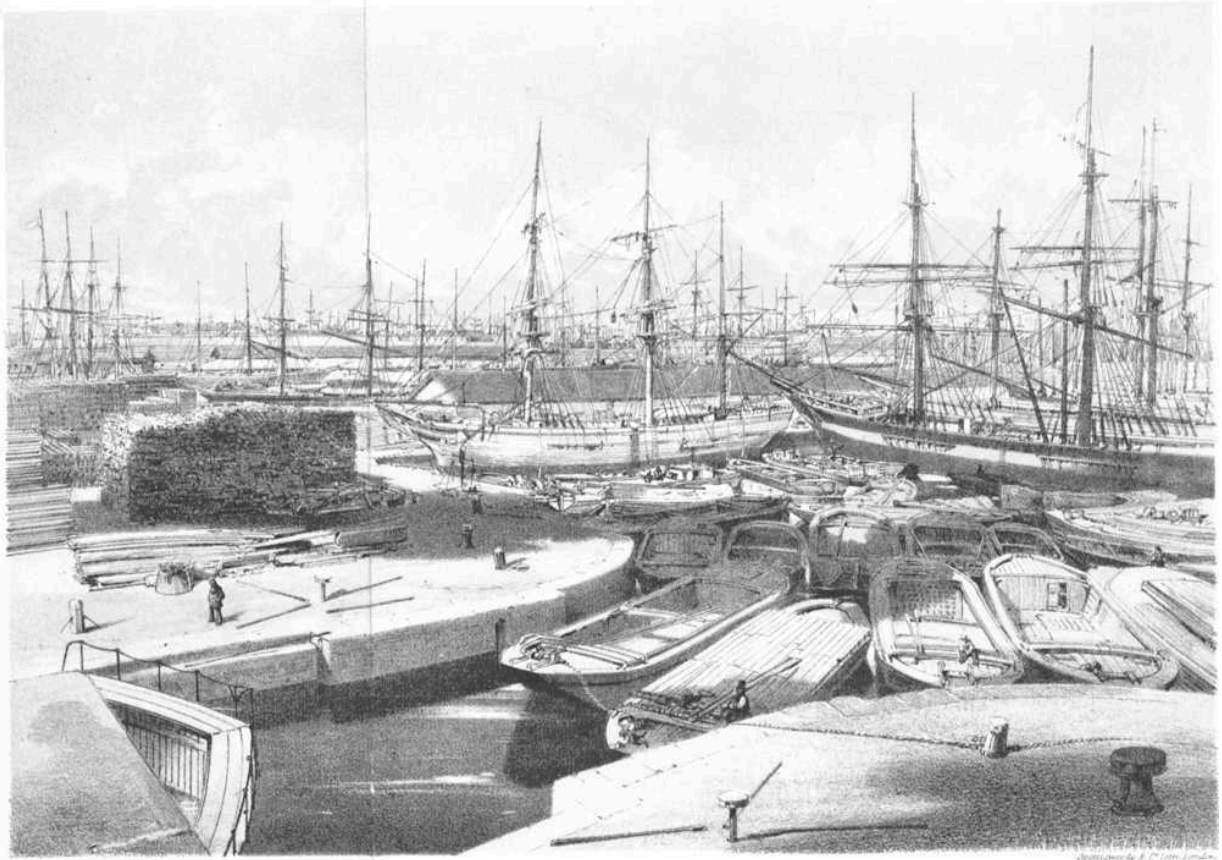
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AND ITS DOCKS ***



SOUTH WEST INDIA DOCK. WESTERN ENTRANCE.

BARGES WAITING TO LEAVE THE DOCK. *See p. [65](#).*

THE THAMES AND ITS DOCKS

A LECTURE

BY

ALEX^R FORROW

(Of the East and West India Dock Company)

DELIVERED AT THE EAST AND WEST INDIA DOCK COMPANY'S LITERARY INSTITUTION,
DECEMBER 18, 1876. CHAIRMAN: PERCIVAL BOSANQUET, ESQ., CHAIRMAN OF THE
EAST AND WEST INDIA DOCK COMPANY.—ALSO AT CITY OF LONDON COLLEGE,
LEADENHALL STREET, FEBRUARY 8, 1877. CHAIRMAN: GEORGE H. CHAMBERS, ESQ.,
CHAIRMAN OF THE LONDON AND ST. KATHARINE DOCKS COMPANY

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TO

J. L. DU PLAT TAYLOR, Esq.

SECRETARY

EAST AND WEST INDIA DOCK COMPANY

THIS LITTLE WORK

IS RESPECTFULLY DEDICATED

BY

THE WRITER

PREFACE.

At the urgent request of many who have heard this lecture, I have been induced to publish it. I have had considerable hesitation in doing so, because, as its structure sufficiently indicates, it was never written with a view to publication. It is a sketch, and nothing more, of a great subject, condensed into a two hours' address. As, however, it contains some curious information of considerable interest, otherwise difficult of access, it has been considered worthy of preservation in its present form. With a view to placing the readers of this lecture, as far as possible, in the position of hearers of it, I have accompanied it with some of the rarest and most interesting of the plans with which it was illustrated. Whatever may be its shortcomings in other respects—and I am conscious they are many—I venture to hope that this feature of it will prove generally acceptable.

A. FORROW.

WOODFORD:
February 1877.

THE THAMES AND ITS DOCKS.

The subject of this evening's lecture is so vast and discursive, that to attempt, within the brief period of a lecture, to give more than a sketch of a particular portion of it, would be manifestly absurd. Moreover, as a visit to any of the Docks will result in the attainment of more practical information than could be imparted in a dozen lectures, we may very consistently dispense almost entirely with reference to contemporary dock history; and I intend doing so except in so far as incidental reference to that part of the subject may appear necessary. My object, in this lecture, is to convey some interesting information respecting the origin and growth of the facilities which have been provided in the river, from time to time, to meet the requirements of the mercantile marine of the port; to trace the steps by which its commerce has gradually been emancipated from its ancient river-side restrictions, and then lead up to notice the establishment of the magnificent system of docks, to which it so largely owes its marvellous growth. I propose to deal with this portion of the subject in preference to any other, because it is one of which, although exceedingly interesting, the literature is singularly scarce, fragmentary, and difficult of access. It is, however, obvious that the ground which we have to cover necessitates such rapid travelling, that it will be necessary to omit many interesting particulars to which I should, otherwise, have invited your attention.

Leaving that misty period of English history when the only craft to be seen on the Thames was the rude coracle of the ancient Briton; and the 400 years during which the Romans occupied the island, I would pass on to observe, that up to the tenth century, there appears to have been so little disposition on the part of our forefathers to engage in maritime pursuits, that in 938, King Athelstane, in order to excite a spirit of enterprise amongst his subjects, offered the rank of thane to any Englishman who undertook three voyages to the Continent with a vessel and cargo of his own. But our real masters in the arts of commerce were the Easterlings, a fraternity of German merchants, who, in the eighth century, established themselves on the shores of the Baltic for the protection of their commerce against the Normans and Scandinavian nations. Some of the vessels of this confederacy, which was afterwards connected with the Hanseatic League, appeared in the Thames at Billingsgate in 979, during the reign of Ethelred

II. Here I should observe that Billingsgate was the first place appointed for payment of king's dues, Queenhithe being the next. Considered by the king worthy of 'good laws,' the Easterlings agreed to pay a toll of two grey cloths and one brown one, ten pounds of pepper, five pairs of gloves, and two measures of vinegar, every Christmas and Easter, and forthwith established themselves in London. Hard-headed, keen-witted fellows were these old German merchants, and, like their countrymen of a later date, soon found out that England was a rare place for making money. And when these ancestors of our modern German competitors first came to this country, they had to encounter so little opposition from the native traders, that they speedily monopolised the whole of the foreign trade of the port. But while the Hanseatic League owed much of its power and influence to the early footing which the Easterlings obtained in London, there cannot be a doubt that by helping to develop the enormous productive resources of the country, and revealing the unrivalled natural facilities of the port for trade, they laid the foundation of the supremacy which London now enjoys as the great entrepôt for the produce and commerce of the world. Moreover, as their remarkable confederacy at one time comprised no less than eighty of the largest and wealthiest cities in Germany and Sweden, their establishment in London was the means of opening up commercial relations with the Continent of a magnitude, which, while the balance of immediate profit fell to the traders of the League, proved an incalculable boon to this country when their exclusive privileges were taken from them. Not only so; the success of the Easterlings in London attracted the commerce of other continental nations, particularly the Venetian and Genoese, and thus brought about the trade with the Mediterranean ports, and, incidentally, with the East. In the reign of Henry III. (1236) the foreign trade had become so important, that the practice which had hitherto prevailed of buying goods on board the importing ship, was found to be so irksome and inconvenient, that, on condition of paying certain tolls to the Mayor, permission was granted for the direct landing of cargo, so that it might be disposed of on shore. This gave rise to the existing landing and wharfage dues. Ten years afterwards, the Corporation purchased Queenhithe, which, it should be remembered, besides Billingsgate, was the only place in the river at that time suitable for the landing of goods. Meanwhile, the enormous influence which the Easterlings had acquired in London, by means of the charters granted them by successive monarchs, was beginning to excite a spirit of

competition and jealousy amongst the citizens. But they found that, great as was the productive resources of the country, their German competitors had obtained such complete control of the continental markets, that no Englishman stood a chance against the combinations which he had to encounter there. This led to counter-combinations on the part of the citizens, as the only means by which they could hope to introduce their goods into foreign markets. I need hardly observe that this was one of the causes which led to the origin of many of the trade guilds, and livery companies of London. In the time of King Edward IV. a considerable impetus was given to commercial transactions of all kinds, and the trade of the port rapidly increased. Troubles on the Continent brought large numbers of skilled artisans to this country, and the result was that, in course of time, the English manufacturers found themselves, as regards the actual cost of production, able to compete with their opponents. Conscious of their growing wealth and influence, the citizens began to raise outcries against their grievances. This feeling sometimes found expression in great popular riots. During one of these outbreaks, which occurred in the year 1493, the premises of the Easterlings in Thames Street, then, and long after, known as the Steelyard, were attacked by the mob; immense damage was done, and the tumult only quelled by the Lord Mayor and aldermen, with all the force they could muster, coming to the protection of the merchants. Some idea of the magnitude of the influence gained by the Easterlings may be gathered from the fact that, for several hundreds of years, they not only determined the weights and measures to be legally used throughout England, but so late as the year 1531, King Henry VIII. granted them a charter exempting them from payment of all king's taxes within the City. But from this period the increased wealth, and growing intelligence of the citizens of London, made it manifest that it would shortly be necessary to withdraw all these privileges. Thus, in the year 1552, the English merchants presented a petition to the king, complaining that their competitors, by trading in a body, kept down the price of woollen cloths (one of the staples of English manufacture), and having command of the foreign markets, shut English merchants out of them. In proof of this statement they alleged, that in the previous year (1551), the Steelyard merchants had exported 44,000 woollen cloths, whereas the English merchants had been able to export only 1,100. It was also represented that the whole of the trade was carried on in foreign bottoms, greatly to the detriment of the English marine. This effort was

successful. The greater part of the monopolies enjoyed by the Steelyard merchants were abolished, and the whole of the trade with Flanders immediately passed into the hands of the English, who, in the same year, exported no less than 40,000 woollen cloths, as compared with 1,100, in the previous year. In the year 1597, all the exclusive privileges of the League in England ceased to exist.

From this period the commerce not only of London, but of England itself, may be said to date its rise. The failure of Sir Hugh Willoughby to discover a North-east Passage led, in 1553, to the formation of the Russia Company, and the opening up of the important trade with Russia and the Baltic; and the stories of the fabulous wealth to be acquired on the newly-discovered continent of America, excited a spirit of enterprise that gave a tremendous impetus to commercial transactions of all kinds. It is quite unnecessary to speak of the encouragement which good Queen Bess gave to this movement, or how far the success of those old sea-dogs, Hawkins, and Drake, and Frobisher, to say nothing of the exploits of the noble Raleigh, contributed to this impetus. Billingsgate and Queenhithe, the two places which had hitherto served for the landing of goods and the collection of the king's Customs, soon proved to be inadequate, and permission was given for the landing of goods at other spots on the river. But this naturally gave rise to attempts to evade payment of Customs dues by smuggling goods ashore at unauthorised places. To repress this practice, an Act of Parliament was passed in 1559, enacting, under pain of very heavy penalties, that all goods should be landed and shipped between sunrise and sunset, and that the landing and shipment was to take place at such open spots as might be authorised by Royal Commission. This Commission, which was appointed by the Court of Exchequer, authorised twenty-two places for the landing of goods. As some of these old landing-places are still known as Legal Quays, it may be interesting to some of you to know that these ancient substitutes for docks were known as:

Old Wool Quay.	Botolph Wharf.
New Wool Quay.	Sab's Quay.
Galley Quay.	Young's Quay.
Androw Morris' Quay.	Crown Quay.
Ambro Thurston's Quay.	Smart's Quay.
Ranff's Quay.	Fresh Quay.

Cock's Quay.	Gaunt's Quay.
Dyce Quay.	Billingsgate.
Bear Quay.	The Three Cranes.
Somer's Quay.	Johnston's Quay.
Busher's Wharf.	The Bridge House.

Billingsgate was appointed only for Fish, Corn, Salt, and Fruit. The Three Cranes and Johnston's Quay for Wine and Oil. Busher's Wharf for Pitch, Tar, and Flax. The Bridge House for Corn and other provisions. Wool, Coals, and Beer might be landed at any place in the presence of a Searcher.

But the rapidly-increasing trade of the port soon outstripped the landing facilities afforded by these places. This led to the laying out of other frontages to the river by private enterprise for the landing of certain articles of merchandise, mostly of a bulky nature, which the Legal Quays gradually became unable to accommodate. These places were known as Sufferance Wharves, as distinguished from the Legal Quays, because for every particular article landed at them it was necessary to obtain from the Customs a special sufferance or permission for such landing, which might at any time be withdrawn; whereas at the Legal Quays any goods might be landed free from any limitation of privilege. As the commerce of the port increased the Sufferance Wharves multiplied with marvellous rapidity, and extended so far down the river as Blackwall. In the year 1590 we find that the total amount of Customs Receipts for England and Wales on Exports and Imports was £50,000. In 1613 they realised £148,075. 7s. 8d., of which London alone contributed £109,572. 18s. 6d. Indeed, so late as 1796 three-fifths of the commerce of the country was centered in London; and, until the commencement of the present century four-fifths of the Customs Dues were also collected here. The ravages of the Great Fire of 1666 necessitated the rearrangement of the Legal Quays, and an Act was passed appointing commissioners to define the limits of the port, and to appoint places for landing and loading goods therein. The wharves set out by the commissioners were twenty-one in number, as follows:

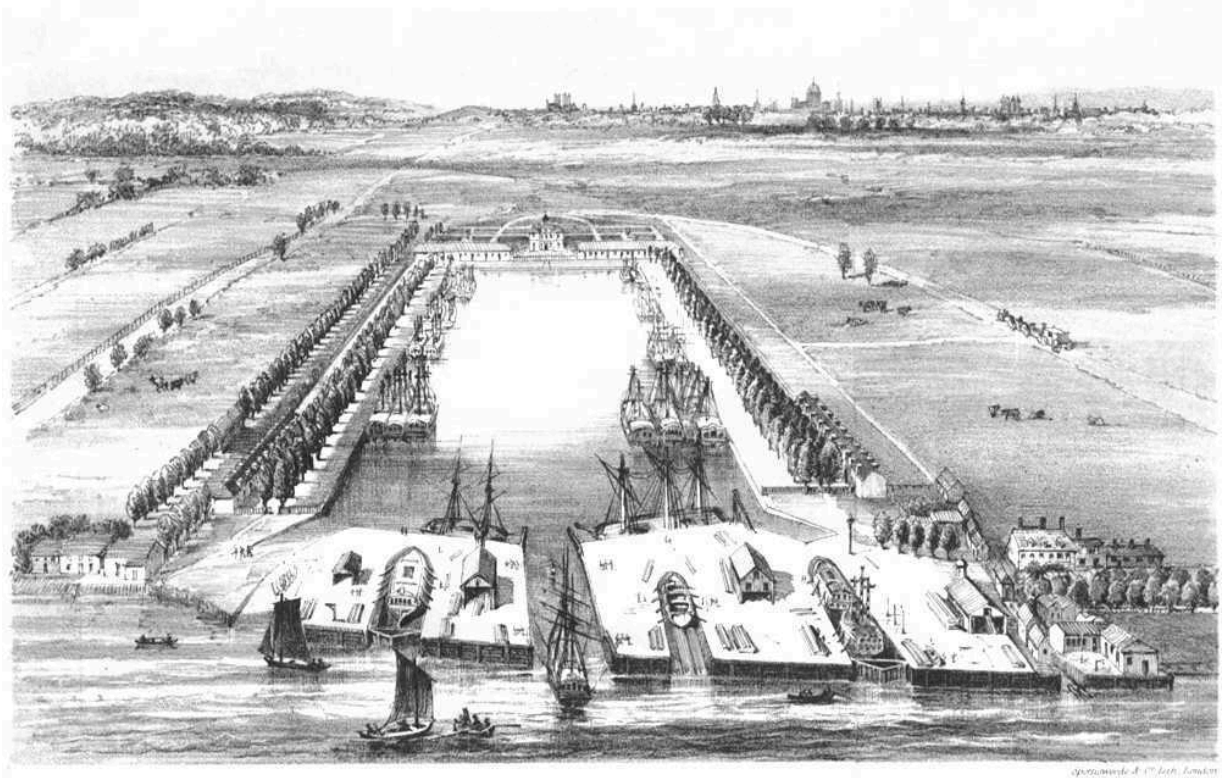
Chester's Quay.	Dice Quay.
Brewer's Quay.	Smart's Quay.
Galley Quay.	Somer's Quay.
Wool Dock.	Lyon's Quay.

Custom-house Quay.	Botolph Quay.
Porteus' Quay.	Hammond's Quay.
Bear Quay.	Gaunt's Quay.
Sab's Quay.	Cock's Quay.
Wiggan's Quay.	Fresh Wharf.
Young's Quay.	Billingsgate.
Ralph's Quay.	

It will be seen that this list includes several of those selected by the earlier commission, and comprised a total river frontage of 1,464 feet only, the whole of it lying between the Tower and London Bridge. As might be expected, this limited accommodation practically left the importer of goods entirely at the mercy of the proprietors of these wharves, because there was no legal limit to the charges to be imposed, similar to the restriction placed upon the Dock Companies when they started. The result was that, as the business of the port increased without any proportionate increase in the accommodation for it, so the wharfingers raised their charges. And they not only did this, but by acting in concert they contrived to evade their responsibilities for losses inseparable from the limited and imperfect nature of the accommodation which they afforded. It was a common complaint amongst the merchants that, owing to the carelessness of the servants of the wharves, lighters were sunk when alongside, goods stolen from the lighters or off the quays; that losses by fire, &c., &c., constantly occurred; and that as the associated wharves were not a corporate body, the parties aggrieved never knew where to look for redress. Now, when it is remembered that these complaints were the subject of a petition from the merchants of London to the Privy Council so early as 1674, and that from that period until the opening of the West India Dock in 1802, not an additional foot of Legal Quay accommodation had been provided, some impression may be obtained of what our commerce must have suffered in the interval. It cannot be doubted that, but for the unrivalled natural facilities of the river for trade, London must have sunk into a third or fourth-rate port. And that such a state of things should have been so long tolerated by London, while Bristol, and Hull, and Liverpool supplied three or four times the extent of Legal Quay accommodation, is all the more surprising. So unsparing were the exactions of the earlier wharf proprietary, that it was ascertained on enquiry that they had trebled their charges in the course of eight years; and justified this

proceeding by alleging it to be the only means by which they could repair the losses which they had sustained by the Great Fire.

Now it is commonly supposed that there were no docks in London in existence at this time. Nor were there on the north side of the river, where they were most wanted, and where the City interest always attracted the landing of the most valuable classes of merchandise. But for many years prior to the date to which I have brought my lecture a most interesting dock had been in existence at Rotherhithe, known as the Howland Great Wet Dock. Stowe tells us that when Canute laid siege to London he commenced to dig a canal on the site of this dock in order to divert the course of the river to Battersea; and in 1209 the current was so diverted to admit of the rebuilding of London Bridge, which in 1176 had been destroyed by fire; and the large opening thus made from the river is said to have formed the nucleus of the dock before you. But, be this as it may, the dock actually existed in 1660 as here represented, and may fairly claim to be the first public dock in Great Britain. As shown in this view, the dock was 1,070 feet from east to west, 500 feet in width, and had a depth of water of 17 feet. It will be observed that on the north and south sides it is thickly planted with trees, the object being to protect the ships in the dock from the fierce gales which in winter swept over the open country surrounding it. On the development of the Greenland Fisheries the dock was specially laid out for the ships engaged in that trade, and was for many years known as the Greenland Dock. Extensive premises, with the necessary boilers and tanks, were erected for boiling the blubber and extracting the oil; and for many years upwards of a thousand tuns of oil were produced annually. At the commencement of the present century this profitable business not only rapidly sunk, and finally left the port of London, but the United Kingdom itself. But simultaneously with the decline of the Greenland trade the timber and corn trade with the Baltic ports increased, and the dock became the principal resort of vessels engaged in them.



HOWLAND GREAT WET DOCK.

IN THE PARISH OF ROTHERHITHE.

FORMERLY BELONGING TO MRS HOWLAND OF STREATHAM

About the year 1807, the period of the inception of the Commercial Dock Company, a Mr. Moore, who owned some forty-five acres of land in the neighbourhood of this dock, projected a Company to be known as the Baltic Dock Company, and succeeded in obtaining from the Treasury a promise of an exclusive right to bond timber in the docks he proposed to construct, provided that they furnished sufficient accommodation for the business. Strange to say, this concession was granted by the Treasury and even notified to the Customs for their information, although the proposed docks were never even commenced. Mr. Moore subsequently sold his land and transferred the exclusive privileges he had acquired to the promoters of what was afterwards known as the Commercial Dock Company.

The City interest always opposed the recognition of this dock as a Legal Quay, for a reason sufficiently obvious, and it was not until 1851 that the

Commercial Dock Company obtained an Act empowering them to land here nearly every description of goods, if sanctioned by the Customs. But the development of the timber trade pointed to the necessity of providing accommodation for it, and the owners of this dock, which in the year 1808 received the name of the Commercial Dock, encouraged by the prospects of the restriction acquired from the Treasury through Mr. Moore, which, as you are aware, was never practically operative, considerably enlarged and improved it; large tracts of land adjoining were purchased and converted into timber ponds, and in 1810 their first Act of Parliament was obtained by the Commercial Dock Company. Granaries, wharves, &c., were subsequently built, and between the years 1810 and 1815 four additional ponds, or docks, were opened. You will thus observe that the Howland Dock was the nucleus of the existing Surrey Commercial system of Docks—a plan of which is now before you. In 1811, the year after the Commercial Dock Company obtained their Act, an Act was obtained for completing and maintaining what was known as the East Country Dock, the history of which seems very obscure. Under their Act of 1851 the Commercial Dock Company purchased this dock. Meanwhile, side by side with the Commercial Docks, the Grand Surrey Canal and Docks had been steadily growing. In the year 1800, Dodds, an eminent engineer, had recommended the Greenland Dock as a suitable entrance to a tidal canal for ships to Vauxhall and Lambeth, along the line now partly occupied by the Surrey Canal. This scheme was never carried out, but the expense and danger of the river navigation indicated the desirability of some similar means of reaching points above bridge by water; and the result was the Grand Surrey Canal, commenced in the year 1800. It was intended to make a cut from Rotherhithe to Battersea, with collateral cuts to Croydon, Merton, Tooting, Wandsworth, and Camberwell. Although empowered by their Act of Parliament to build docks, the Surrey Company did not exist as a Dock Company until 1854. Ten years afterwards this Company united with the Commercial Company, and thus formed, under the title of the Surrey Commercial Docks Company, the interesting and extensive network of water-communication which you see before you. Including the new dock recently opened, the total extent of these docks—land and water—is nearly 370 acres.

Now returning to the north side of the river, I would ask you to bear in mind my reference to the petition of the London merchants in 1674. But

anything like redress from a Privy Council of which King Charles II. was the president, was by no means easy of attainment. The king wanted money. As is well known, his favours generally went to the highest bidder. Moreover, the wharf proprietary consisted for the most part of the leading City magnates, and to conciliate them—or, rather, not to do anything to offend them—was obviously a great point with this most worthless of merry monarchs. So the merchants petitioned in vain. But in spite of these serious drawbacks the trade of the port steadily increased. An immense impetus was given to the manufacturing industries of the country by the revocation of the Edict of Nantes in 1685. This measure had the effect of sending from 50,000 to 70,000 of the most skilled of the French artisans to this country, and some £10,000,000 of French money. The result was increased activity in the manufactures of the country, and a great accession in the trade of the port. Thus, while in the year 1613 the total value of the exports and imports had been a little over £4,800,000, in the year 1700 they represented a value of £13,272,891, of which London alone contributed £10,263,325. It will, of course, be readily understood that in the absence of easy and cheap means of internal communication, the only method by which the manufacturing districts could get their goods into the London markets would be by coasting-vessels. Hence this class of business constituted the bulk of the shipping frequenting the river. Thus, in the year 1728, while 1,839 British and 213 foreign vessels entered the river, the number of coasting-vessels, colliers, &c., was not less than 6,837. At this time most of the foreign trade was in the hands of the East India Company, the Russia Company, the Levant Company, the South Sea Company, the Hudson's Bay Company, and the African Company. With the exception of the cargoes carried by the ships of the East India Company, much of the business conducted by these great companies could be accommodated at the Sufferance Wharves, and did not present the inducements to plunder offered by the valuable productions of the East and West Indies. The conquests of Clive in the East Indies resulted in an immense augmentation of the shipping of the East India Company, and the acquisition of many of the West India Islands about the same time led to a vast increase in the importations of sugar and rum, which, from their value and the high duties to which they were subject, offered tempting baits to the organised bands of river thieves which the crowded state of the port and the unprotected condition of the shipping and merchandise gradually called into existence.

The Legal Quays could accommodate 32,000 hogsheads of sugar only; whereas, so early as 1756, the annual importations reached upwards of 60,200, and over 5,000 puncheons of rum. The Sufferance Wharves could accommodate 60,200, and in the year 1793 the Government were compelled to sanction the landing and storage of sugar at these places, although laid out for, and principally used by, the coasting trade. In the year 1796, the Sufferance Wharves on the south side of the river were known as:

Chamberlayne's,	Hartley's,
Cotton's,	Pearson's,
Haye's,	Holland's,
Beale's,	Cole's,
Griffin's,	Carrington's
Symon's,	Hoggarth's,
Stainton's,	Scott's,
Davis, Butt & Co.'s,	Merriton's,

representing a river tonnage of 2,890 feet. Those on the north side of the river were:

Irongate,	Bryant's,
St. Catherine's,	Down's,
Watson's,	

representing a frontage of 786 feet; or a total for both sides of the river of 3,676 feet.

In many instances, the charges at the Sufferance Wharves were even higher than those at the Legal Quays, to say nothing of the increased liability to plunder incurred by merchants in sending their goods there. And not only was this the case, but as Customs officers were only officially stationed at the Legal Quays, whenever they did duty at the Sufferance Wharves they were paid extra fees at the consignee's expense; and when permission was given to land the whole of a cargo at these places, the consignee of each parcel of goods had to pay extra fees for his particular parcel, and those *pro rata*, in order that a separate sufferance might be taken out for them. If, therefore, after waiting four or six weeks to get his goods landed at one of the Legal Quays, a merchant, in despair, sent them to a Sufferance Wharf, it was like jumping out of the frying-pan into the fire.

Again and again the necessity of extending the Legal Quays was brought under the notice of Parliament, and in 1762 Lord Bute, then Prime-Minister, warmly supported a scheme for thus laying out a large tract of ground, including the Tower Ditch; but with all his influence, the parties interested in the maintenance of the then existing order of things were too strong for him, and the scheme had to be abandoned. Three years later another commission was appointed; but, after a bitter controversy, their proposition was rejected on the frivolous ground that the land which they proposed to acquire and lay out as Legal Quays was not an 'open space' as defined in the Act of Queen Elizabeth. From this time until 1793 various plans were suggested, but all of them were frustrated by the powerful combination of the wharfingers.

Thus the condition of things on the river went on growing from bad to worse, until we reach the years 1793-94, when the crowded condition of the river was declared to be altogether intolerable. Meanwhile, in spite of these drawbacks, the increase in the commerce of the port during the twenty-four years ending with the latter year had been as great as in the first seventy years of the century. In the year 1702, the shipping entered inwards, foreign and British, exclusive of coasting vessels, was 1,335 ships, with a tonnage of 157,035 tons; in 1751 it was 1,682 ships, with a tonnage of 234,639 tons; in 1794 it was 3,663 ships, with a tonnage of 620,845 tons. From a little over £10,000,000 in the year 1700, the value of the exports and imports had rushed up to £31,442,040. With Legal Quay accommodation for only 32,000 hogsheads of sugar, the annual importations had reached upwards of 140,000 hogsheads. During war time, the West India fleets only could more than trebly fill the warehouses. One fleet from the Leeward Islands brought 35,000 hogsheads; another from Jamaica brought 40,000: on the arrival of the latter fleet only 7,000 hogsheads could be warehoused at the Legal Quays; the rest had to remain in barges in the river at the mercy of the river thieves, or be left on board the importing vessels until it could be landed elsewhere. I may add that in five months in the year 1794 not less than 122,000 hogsheads of sugar arrived in the port. Indeed, at this period the Legal Quays could not accommodate one-fourth of the trade.

I need hardly remind you that in the absence of quays at which vessels could lay alongside and discharge, their freights had to be unshipped in the river and crafted ashore, and some arrangement existed as to those portions

of the stream in which vessels from different parts took up their position for this purpose. Mooring chains extended from the Upper Pool to Limehouse, on both sides of the river. Each of these mooring chains was intended to afford anchorage for fourteen or sixteen vessels; but, sometimes, as many as thirty were moored to each. The moorings from London Bridge to the Tower were for the most part occupied by coasters; those at Blackwall were for East Indiamen; and West Indiamen and king's ships moored off Deptford. Altogether there was mooring accommodation for 800 vessels, large and small. Now, as I have already shown, in the year 1794, 3,663 ships, British and foreign, with a tonnage of 620,845 tons; and 10,286 coasters, with a tonnage of upwards of 1,000,000 tons, competed for this limited anchorage. Of course, hundreds of vessels grounded with the tide, and in rough weather sometimes sustained more injury than accrued from two or three bad voyages. These figures will help to convey some impression of the crowded state of the river, although I am unable to say what number of vessels were, as a rule, in the river at one time. This condition of things was greatly aggravated by the exigencies of war, which, by throwing ships into large convoys, resulted in the river at times being completely blocked. To increase the confusion it would sometimes happen that as many as 300 colliers would come up the river at one time, and although a regulation existed that a fair way of 300 feet should always be preserved for vessels passing up and down the stream, the crowd of shipping has been so great that one might walk from the Middlesex to the Surrey shore over the decks of vessels. Here I may remark that, in the winter of 1794 coals reached the enormous price of six guineas the chaldron; for although there were hundreds of colliers in the river, the glut of shipping was so excessive that they could not be unloaded. Accidents and loss of life were matters of everyday occurrence: over 500 persons lost their life from drowning annually, three-fifths of these cases occurring in the Pool. But by far the most serious of the many evils arising out of this condition of things was the immorality which it generated amongst all classes of labour employed on the river. East Indiamen seldom came higher up the river than Blackwall, where they discharged their cargoes into decked lighters, or hoys, of from 50 to 100 tons; hence the pillage was, comparatively speaking, trifling. By far the greater bulk of the most costly and valuable merchandise, such as rum, sugar, wine, &c., was discharged into open lighters, punts, billyboys, lugger-boats; and, to say nothing of the

plunder effected during discharge, and in the passage up the river to the wharf, these craft unprotected had to lie up, sometimes as long as six or eight weeks, before they could be unloaded, not a night passing without sustaining loss of some kind. Lightermen, labourers, watermen, the crews of ships—in some instances recorded, the officers included, and even the Revenue officers—combined in this nefarious system. The sides of the river swarmed with receiving-houses, many of them kept by persons of considerable opulence; and when it is borne in mind that the value of property floating on the river was estimated at £70,000,000 per annum, it will be seen what a rich harvest these rascals had to reap. The men engaged in the discharge of ships were under little or no control, and their depredations had come to be of so systematic a character that they were regarded as one of the taxes of the port. The several classes of thieves were even known by special designations to indicate the field of their operations. The body known as the ‘River Pirates’ were the most formidable of these marauders. Most of them were in open league with the marine store dealers. Reconnoitring by day, they made their attacks upon ships by night in armed boats, cutting adrift lighters and taking out their contents. The ‘Night Plunderers’—watermen of the lowest class—attacked unprotected lighters and made over the booty to the receivers. The ‘Light Horsemen’ comprised the mates of ships and Revenue officers. Then there were the ‘Heavy Horsemen,’ consisting of the porters and labourers employed on board ships, who wore dresses specially made with big inside pockets to secrete anything of value that was not too heavy to carry off; to say nothing of the ‘Mud Larks,’ who picked out of the mud on the shore property thrown overboard by persons in concert with them at work in the ship. Some of these young thieves are said to have received as much as £5 a night for their booty. The losses sustained in wines and spirits were something enormous. Instances are recorded of lighters with double bottoms being employed; and by dexterously loosening two or three hoops of casks and turning them bung downward, hundreds of gallons of rum, wine, or oil—as the case might be—were deposited in the false bottom during transit from the ship to the wharf. The greatest sufferers were the West India merchants. Mr. Hibbert, one of the most eminent of our West India merchants, speaking in 1822 of this condition of things, gives a graphic description of the grievances of the trade. Both rum and sugar samples were drawn on board ship. The aggregate weight of the sugar samples drawn from each cask

averaged 12 lbs. In breaking out hogsheads of sugar the casks were often knocked to pieces, in order that their contents might be picked up as scrapings, and claimed as perquisites. Sometimes whole packages were appropriated. Of a cargo of 120 puncheons of rum brought by one vessel 7 puncheons were thus *missing*. Mr. Lindsay, in his admirable work, 'A History of Merchant Shipping,' gives some interesting details of the method in which these transactions were conducted. He says:

'Most of these infamous proceedings were carried on according to a regular system, and in gangs, frequently composed of one or more receivers, together with coopers, watermen, and lumpers, who were all necessary in their different occupations to the accomplishment of the general design of wholesale plunder. They went on board the merchant vessel completely prepared with iron crows, adzes, and other implements to open and again head up the casks; with shovels to take out the sugar, and a number of bags made to contain 100 lbs. each. These bags went by the name of "black strap," having been previously dyed black to prevent their being conspicuous in the night, when stowed in the bottom of a river boat or wherry. In the course of judicial proceedings, it has been shown that in the progress of the delivery of a large ship's cargo about ten to fifteen tons of sugar were on an average removed in these nocturnal expeditions, exclusive of what had been obtained by the lumpers during the day, which was frequently excessive and almost uncontrolled whenever night plunder had occurred. This indulgence was generally insisted on and granted to lumpers to prevent their making discoveries of what they called the "drum hogsheads" found in the hold on going to work in the morning, by which were understood hogsheads out of which from one-sixth to one-fourth of the contents had been stolen the night preceding. In this manner one gang of plunderers was compelled to purchase the connivance of another, to the ruinous loss of the merchant.'

Thousands of persons were employed simply to *watch* goods until landed. Of 37,000 persons employed on the river, 11,000 were either professional thieves or receivers of stolen property. Over 1,600 Customs officers were employed on the river in the interest of the Revenue. To an East Indiaman of 400 or 500 tons thirty officers were attached, and to a West Indiaman from seven to ten. While on board these officers were generally fed at the expense of the ship. The value of property annually

stolen on the river was upwards of £500,000, and involved a further loss to the Revenue of £300,000. The Marine Police—as the Thames Police were formerly called—in the first year of their existence recovered stolen property worth upwards of £100,000. So bitter was the feeling amongst the river thieves against the magistrates and members of this useful body when established in 1797, that they made a most daring attack upon the office, with a view to intimidation. In defending themselves the magistrates were compelled to use fire-arms. Several persons were killed, and a number injured in the fray. As a further illustration of the delays incurred under this condition of things, I may add that goods were often seized by the Customs though duly entered, and in some cases, even the duties paid. Wines were required to be landed within twenty-one days; rum and coffee were allowed thirty days; whereas sometimes vessels remained in the river two months before they entirely got rid of their cargoes. On one occasion 5,000 puncheons of rum in this way became subject to seizure, and would have been seized had not the Customs granted a special indulgence.

Before proceeding to call your attention to the various schemes which at this time (1794-96) were submitted to the Government for the improvement of the port, I wish to bring under notice the existence of a small private dock at Blackwall. This dock, commenced on March 2, 1789, and known as the Brunswick Dock in honour of the reigning royal family, was constructed by Mr. Perry, of Messrs. Perry, Wells, & Co., who owned the adjoining ship-building premises, and opened on November 20, 1790. Here I may observe that, from a very early period, Blackwall has been a noted place for ship-building yards. Pepys speaks of a visit which he paid to them in 1661, and again in 1665, when, much to his discomfort, he complains that he was compelled to pass a night in what he calls the ‘unlucky Isle of Doggs.’ I may also mention in passing that Mr. Perry built a small dock in 1783, just large enough to receive one whaling vessel. This little dock, of which I have not been able to obtain a view, was subsequently converted into one of the slips now occupied by Messrs. Wigram’s dry docks. The Brunswick Dock was built for the accommodation of the vessels of the East India Company, and was capable of receiving twenty-eight large Indiamen and a number of smaller vessels. This dock formed the nucleus of the existing East India Export Dock, and many of you remember the old Mast House, which, you will observe from its appearance here, was coeval with the origin of the existing dock. But the relief afforded by this dock was merely a drop in the

bucket, and from the years 1793 to 1799, when the Bill for the construction of the West India Dock was passed, we find a number of schemes for increasing the accommodation of the port engaging the attention of the mercantile community and Parliament. These, I now proceed, as briefly as possible, to notice in the order in which they were reported upon by the Committee of the House of Commons appointed to take evidence and consider and report on them.

The first of these schemes, to which I would invite your attention, is what was known as Mr. Ogle's scheme. Mr. Ogle was a wealthy ship broker and ship's-husband. He was also Chairman of the Committee of Proprietors, Lessees, and Wharfingers of the Legal Quays, and, without doubt, one of the most intelligent and able men of his party. It is a fact, which should always be borne in mind, that the scheme submitted by this gentleman—who was, of course, deeply interested in getting it adopted—was the only plan before the public based on the theory that the river could be so deepened, and otherwise improved, as to answer the purposes of the commerce of the port. *All the other plans were founded on the one general principle of the necessity of docks or recesses being made out of, and clear from, the channel of the river.* I lay particular stress upon this fact, because it is the practice in some quarters to regard the origin of the docks of London as exhibiting something like an act of grace on the part of the wharf proprietors, and only tolerated by them on conditions from which the Dock Companies in a spirit of doubtful faith would now gladly be absolved. But I think I have already said enough to prove that no view of the question could be less in harmony with the facts. The docks of London were started, in the first instance, not so much as a profitable speculation to investors, as a response to an urgent national demand which the vested interests in the old order of things were no longer able to resist; not so much in a spirit of competition with the wharves, as the only means of saving the commerce of the port from utter ruin; not so much as wealthy grasping corporations, who, finding they have made a bad bargain, now wish to draw out of it, as the pioneers of the marvellous prosperity of the port, if not its founders, its liberators. Certainly, without its docks, the London of to-day would have been an impossibility; but of this I shall speak more fully presently.

Mr. Ogle, whose plan is before you (see Appendix [A](#)), proposed to deepen the river, extend and improve the Legal Quays, and to increase the

number of mooring chains. He further proposed that the moorings should be appropriated in proportion to the claims of the several classes of commerce, and that harbour masters should be appointed to see that vessels took up their proper position as they came up the river. The different colours on the plan represent the moorings. It was, however, the opinion of nine-tenths of the practical men of the port that under no scheme would it be possible so to improve the river and extend the Legal Quays as to obviate the necessity for wet docks.

The next scheme to which I would ask your attention was known as the Merchants' Plan of Docks at Wapping (see Appendix [B](#)). The desirability of having goods landed as near the City as possible is, of course, obvious. As a natural consequence, all the schemes for docks at Wapping were warmly supported by the City. The dock here represented was to be 39 acres in extent, with a smaller one of 2 acres for the accommodation of lighters. One of the entrances was to be by a canal 22 feet deep, 170 feet wide, and $2\frac{3}{4}$ miles in length, and navigable for ships of 300 tons: and, as you will see, communicating with the river immediately above Perry's Dock, at Blackwall. The object of this canal was, of course, to avoid the circuitous and dangerous navigation round the Isle of Dogs, which was always a great difficulty with the early dock engineers. To get from the Pool to Blackwall when 'kedging' was the only means of getting a vessel along, sometimes occupied fourteen days, and it was felt that no scheme would be satisfactory which did not in some way enable vessels to avoid this obstinate bend in the river. The line taken by this canal was, as nearly as possible, that now occupied by the Commercial Road. This scheme was estimated to cost £1,000,000. While this plan is before you, I wish to call your attention to a small canal known as the 'Bromley Cut,' connecting the River Lea with the Thames at Limehouse. It is not generally known that this Cut was commenced so early as the reign of Elizabeth, in the year 1571. Before this Cut was made, the connection of the traffic on the Lea with the City was most difficult, inasmuch as barges had to leave the mouth of that river at Blackwall, and wind their way round the Isle of Dogs—a most hazardous undertaking. All this was obviated by the now despised and unsavoury Cut. This system of docks, although not accepted, was the basis of the existing London Docks.

The third plan was known as the ‘Corporation Scheme’ (see Appendix [C](#)). This scheme, with the munificence which distinguishes every undertaking of the Corporation of London, was at once the most extensive and expensive of the many schemes which engaged public attention. The Corporation proposed to excavate one dock in the Isle of Dogs of 102 acres, and another at Rotherhithe of the same extent, with a canal to Vauxhall. They further proposed to extend the frontage of the Legal Quays by the acquisition of Billingsgate, and to provide for the construction of slips for the accommodation of lighters. They also intended largely to increase the warehouse space, by arching over the quays and constructing warehouses on them.

This plan (see Appendix [D](#)) represents what was known as Mr. Wyatt’s Scheme. Mr. Wyatt was an eminent architect and civil engineer. His plan of docks in the Isle of Dogs was, as you will observe, very extensive—so extensive that, although the price of land in the island was then only £5 per acre—indeed the whole of the island might have been purchased for £10,000—and labour was cheap, the estimated outlay was nearly £900,000. The water space afforded by the docks and basins proposed by Mr. Wyatt would have been upwards of 200 acres; and in view of the vast increase in the commerce of the port, and the present great cost of dock extensions, one cannot but regret that some modification of this scheme was not adopted.

This plan of docks at Rotherhithe (see Appendix [E](#)), designed by Mr. Cracklow, a surveyor, and which bears some slight resemblance to the network of docks constituting the Surrey Commercial system, was known as the ‘Southwark Plan,’ and included a canal opening to the Thames at Bankside, above London Bridge. These docks, principally intended for colliers, timber ships, and vessels for sale, were the most inexpensive of the many schemes proposed, the estimated outlay being only £300,000.

These docks at Wapping (see Appendix [F](#)) were proposed by Mr. Walker, a wealthy shipmaster and Jamaica planter. The docks here shown included a water space of 55 acres, leaving 35 acres for quays and warehouses. You will observe that Mr. Walker also proposed a canal similar to that suggested by the Merchants’ Plan, but taking lower ground near the river. Besides the access to this dock afforded by the canal, Mr. Walker proposed two others: one at a little bay known as ‘Hermitage Dock,’ and the second at a place called ‘Pellican Stairs.’ His scheme also included a dock in the Isle of Dogs

for timber-ships, reaching across the island by what was known as ‘Poplar Gut,’ a large piece of swamp or boggy ground at the Limehouse side of the island, and now absorbed by the West India Docks. The total cost of this great scheme was estimated at £880,000. This plan of docks was very warmly supported by the Brethren of the Trinity House, who alleged that it furnished every requisite for the accommodation of the trade of the port and the proper navigation of the river; and I am sure that those of my hearers who would like to see a dock of 55 acres at Wapping, will agree with me in expressing regret that this scheme was not carried out.

This plan (see Appendix [G](#)) represents a scheme for docks proposed by Mr. Spence, a maritime surveyor to the Admiralty. He proposed to divide the shipping of the port into twelve different classes, each class to have a separate dock for its accommodation. He proposed that six of these docks should be 600 feet square, the other six being 400 feet, connected with each other on the plan of the docks at Liverpool. As you will see by this plan, Mr. Spence suggested two alternative sites for these docks. There cannot be a doubt that, had the principle here indicated of localising the warehousing of the various classes of merchandise been more fully carried out, the working of goods would have been more efficient, more economical, and more satisfactory in every way to both merchant and Dock Company. In the monopolies granted to the earlier Dock Companies, the Government seem to have made an attempt in this direction; but it is to be feared that the spirit of competition and the morbid jealousy of anything tending to a monopoly will always be a barrier to any extended application of this principle, however much it might be the interest of the public to promote it. The docks proposed by Mr. Spence were estimated to cost £500,000.

The last, but by no means the least interesting, of these rejected dock schemes to which I shall invite your attention, consists of four alternative plans proposed by Mr. Willey Reveley, an engineer and architect. You will observe by the plan before you (1), that Mr. Reveley proposed (see Appendix [H](#)), by a bold stroke, at once to demolish the Isle of Dogs, as an impediment to the navigation of the river, by cutting a channel straight through it from Limehouse to Blackwall; leaving the long reach round the island as a magnificent dock of 434 acres, with flood-gates at each entrance to the new course of the river.

Under the second scheme proposed by him (see Appendix [I](#)), Mr. Reveley, as you will observe by this plan, suggested the cutting of a new channel for the river, inclining towards Woolwich Reach below Blackwall, so as to convert the upper bend of the river by Perry's Dock into a second dock, thus securing for the two docks a water space of 524 acres.

Under his third plan (see Appendix [J](#)), Mr. Reveley proposed to conduct the new channel of the river straight from Wapping, intersecting the river so as to convert the three bends between Wapping and Woolwich into three docks, to be known respectively as the 'Ratcliff,' the 'Blackwall,' and the 'Greenwich' Docks, giving a total of 644 acres.

To meet the objection of the Trinity House that any of these schemes would be impracticable without essential injury to the river and its navigation, Mr. Reveley proposed a fourth plan (see Appendix [K](#)), under which, as you will see, the new channel of the river is made to take its course from near Wapping to the old channel of the Thames between Greenland Dock and Deptford, thence inclining gently to the northward till it falls into Woolwich Reach; thus leaving two spacious docks to the northward by shutting out the Ratcliff and Blackwall bends of the river. The docks thus formed would have yielded a water space of 559 acres.

Each of these rejected schemes, of course, represented particular, and, in some instances, conflicting, interests. The West India merchants who suffered most severely from the depredations on the river, were very anxious to have a dock in the Isle of Dogs, and in two days, December 22 and 23, 1795, raised subscriptions of £800,000 for the purpose. Opposed to the West India merchants was the Corporation of London, with its large collateral vested interests. They professed to be jealous of any measure which would have the effect of removing the shipping from the City; hence their gigantic scheme, to which I have called attention, and which, as you will remember, so far met the views of the West India merchants as to include a large dock in the Isle of Dogs. The fact is, the Corporation wished to get the control of the docks of London into their own hands; and although Parliament was not inclined in this way to swell the sufficiently plethoric bulk of the Corporation, that body, with its allied interests, was strong enough in Parliament to defeat any body of merchants going to it for powers opposed to their interests; and, as you are aware, all the Dock

Companies when first started were largely represented by nominees of the Corporation.

So the struggle went on for three years longer. Meanwhile a terrible outcry was raised in the City against the proposed establishment of docks of any kind either at Wapping or in the Isle of Dogs. It was urged that the conservation, rule, and government of the Thames was vested by prescription, confirmed by charters and Acts of Parliament, in the Lord Mayor for the time being; and that to embank or inclose the bed of the river, or make any cuts into it without previous license under the City Seal, was against the law. The 'Tackle-house' porters, the 'City' porters, the 'City' carters, watermen, and others, raised a jeremiad of their impending ruin if docks were allowed. One of the objections was that Christ's Hospital would lose £400 a year, granted from the sale of licenses. The proprietors of the Legal Quays protested that the withdrawal of the West India trade alone would lower their property to one-third of its value. It was even said that so many men would be thrown out of employment that a vast increase would take place in the poor rates. One cannot help feeling puzzled to understand how objections of this kind could ever have been seriously advanced. At length, after apparently endless delays, Parliament succeeded in adjusting the conflicting claims of the various parties by payments out of the Consolidated Fund, amounting in the aggregate to £1,600,000; and on July 12, 1799, passed the first Dock Act. Under this Act, which is known as an 'Act for rendering more commodious and better regulating the Port of London,' the Corporation of London were authorised to construct a canal in the Isle of Dogs, with power for fourteen years to levy tolls of from 1*d.* to 3½*d.* per ton on all vessels using the port, fishing and passenger craft excepted. The incorporation of the West India Dock Company immediately followed. The West India Import Dock commenced on February 3, 1800, was completed at a cost of £17,000 per acre, and opened on August 27, 1802, by William Pitt, the Prime Minister. This view represents the appearance of the dock before the water was let in; and, I may add, in passing, that in excavating this dock, as also during the progress of the works at the South Dock, to which I will call your attention presently, some most interesting fossil remains and other curiosities were discovered.

This view represents the first ship entering the dock—the 'Henry Addington,' a West Indiaman of the old school, and one of the wooden

walls, in privateering days, of England's reputation. It is to be hoped that the iron walls which have succeeded them in the Navy will give as good an account of themselves in 'the battle and the breeze' as these old wooden walls; and so dispel the misgivings to which some of their recent achievements have given rise.

This view represents both the West India Import and Export Docks, with the canal, as they appeared when completed in 1805. I should here state that in 1829 the Corporation of London sold the canal to the West India Dock Company for £120,000; and, until converted into the existing South Dock, it was used by the Company as a timber pond, and for the accommodation of an important grain trade. The Government was so impressed with the necessity of removing the shipping from the river, that for a period of twenty-one years they made it compulsory upon all vessels from the West Indies to discharge in these docks. Not only so; but the Commissioners of Customs were empowered to order vessels from other parts to discharge here. And all vessels outward bound for the West Indies were compelled either to take in their cargoes in these docks or else in the river below Blackwall—an arrangement more costly than the payment of dock dues. As a further illustration of the interest which the Government took in these docks, I may add, that the wall surrounding them was built at a cost of £30,000, advanced by the Government; and that for a considerable time the premises were under the protection of troops sent down for the purpose, and that circumstance accounts for the existence of the guard-houses to be seen facing the inner entrance to the West India Docks.

You will have observed that the West India Docks were established, almost exclusively, in the interest of the West India trade. Their location in the Isle of Dogs had, of course, been in opposition to the views of a powerful party in the City. The result of this feeling was the establishment of the London Docks, which followed so closely upon the West India Docks that scarcely twelve months elapsed between the passing of the two Acts; the London Dock Act passing on June 20, 1800. Commenced on June 26, 1802, the first stone being laid by Lord Hawkesbury, Chancellor of the Exchequer, the 'Western' Dock of 20 acres, the first completed, was opened on the last day of January 1805, at a cost £140,654 an acre: and this view represents the dock as it then appeared. This is a remarkably fine view, as

you will observe, the sinuous course of the river and the Isle of Dogs in the distance, coming out very distinctly.

This view represents the dock, when opened, as seen from the river. All vessels entering the port with wine, brandy, tobacco, and rice were compelled to unload here for a period of twenty-one years from the date of completion, under a penalty of forfeiture of the ship to the Crown, and a fine of £100 from the owner or master. The Hermitage Basin and entrance were opened in 1820, the Eastern Dock and the Tobacco Dock in 1828, and the original Shadwell entrance and basin in 1832. The fine jetty in the Western Dock was built in 1838, and the new Shadwell entrance and basin were constructed at a vast outlay and opened on October 13, 1858.

The establishment of the West India and the London Docks still left the East India and China shipping to be provided for; the accommodation afforded by Mr. Perry's Dock at Blackwall being scarcely sufficient for the shipping of the East India Company alone. To meet this deficiency the East India Dock Company was formed. The Act passed in the year 1803, the docks were commenced in August of the same year, and opened on August 4, 1806. This view, which represents the docks as they appeared when completed, gives a very interesting picture of the river at this point, as also of the virgin character of the surrounding country. Like its predecessors, the East India Dock Company started with a twenty-one years' monopoly. All vessels with cargoes from the East Indies and China were obliged to discharge in these docks. Outward-bound ships to these parts of the world were also compelled to load here or else in the river below Limehouse. But, for the protection of the London Dock monopoly, it was enacted that no vessel not immediately from, or immediately bound to, the East Indies or China, should, under a penalty of £50, enter these docks without the consent of the Treasury in writing.

Most people are not aware that the East of London is indebted to the East and West India Docks for two of its finest roads. Before the establishment of these docks neither the Commercial, nor the East India, Road had any existence. The distance of these docks from the City rendered good roads an absolute necessity, especially in those days. Here I may observe that when the Duke of Bridgewater began his canal—forty years previously—the cost of the land carriage of goods was 40s. a ton. The carriage of goods on the river was not less than 12s. a ton. From Wapping, eastward, fields, nursery-

grounds, rope walks, gardens, &c., &c., stretched right away to Barking. Through these the Commercial Road was cut at a cost of £100,000; and when the East India Docks were started, a further sum of £20,000 was raised to continue the road to that distance, the extension taking its name from the docks. The government of both these roads was vested in fifteen trustees, including the Chairman and Deputy Chairman of the two Dock Companies. As you are probably aware, both these roads are now entirely free.

This is a view of the Limehouse Dock as it appeared shortly after it was opened, and before the Blackwall Railway was constructed. This dock was built in connection with the Regent's Canal for the accommodation of lighters, and the Act for its construction was passed so early as the year 1812. Although never, like the other docks, recognised as Legal Quays, this dock has rendered admirable service to the river by relieving it of a large portion of the coasting trade. Everybody who has travelled by the Blackwall Railway must have observed the crowded condition of this dock. Since first opened it has been enlarged three times, and now possesses a water area of 10 acres. In 1869 a ship entrance, 350 feet long and 60 feet wide, was opened, with cills laid 28 feet below Trinity high-water mark.

This view represents the dock from the river. Most of you are aware that an extensive scheme is now on foot for enlarging this dock, and making it a great Railway Dépôt. If the scheme proposed should be carried out, the dock will be enlarged to three times its present size, and the whole character of the neighbourhood will be changed by the alterations necessary.

Time will not permit me to refer to the questions which agitated commercial communities in London on the expiration of the monopolies which had been granted to the early Dock Companies. Suffice it to say that so early as the year 1823 the St. Katharine Dock Company was established chiefly, as I am informed, at the instigation of one of the large trading companies, which had taken offence at the London Dock Company. But this dock was no doubt mainly started as a competitor for the trade that would be liberated on the expiration of the monopolies granted to the other companies. The Government had refused to renew that of the West India Dock Company, which had expired in 1822, and those of the London and East India Dock Companies would expire in 1826 and 1827 respectively. It was thus clear that a considerable portion of the business of the port which

had hitherto been forced into particular docks would soon be free to go where it pleased. These considerations were the primary motives with the originators of the St. Katharine Docks, who, moreover, comprised some of the leading merchants anxious for a reduction of rates. But owing to the opposition of the London Dock Company, into which I cannot now enter, the St. Katharine Company, though started in 1823, could not commence their dock until May 3, 1827. But through extraordinary exertions it was completed and opened on October 25 of the following year, at the enormous cost of £195,640 per acre. The picture before you gives a bird's-eye view of the dock, which comprises a water space of 11 acres. This was the first free dock in London, and its promoters based their appeal to the public for support on the allegation that the charges of the other Dock Companies were exorbitant, and injurious to the interests of the port. Here I may observe that two other Bills for the construction of docks passed in the same year as that for the construction of this dock; one on the south side of the river, and another for colliers in the Isle of Dogs. But both projects were abandoned. In 1825 a ship canal from London to Portsmouth was projected, navigable by vessels of the largest size, and the prospectus, which may be seen at the British Museum, is very interesting.

From the opening of the St. Katharine Dock in 1828, until the passing of the Victoria Dock Act in 1850 (July 15), there was a lull in dock enterprise. But the vast and rapid increase in the commerce of the port appeared then to suggest the desirability of further dock extension; and the Victoria Dock, the largest artificial dock in England—having a water space of 90 acres—was the result. Most of you are probably aware that this magnificent dock is now the property of the London and St. Katharine's Company, and as its history previous to the purchase of it by that Company is of such recent date, it is quite unnecessary for me to refer to it. In a short time this grand dock will be extended by a ship canal to Gallion's Reach; and this view represents the dock and canal as they will appear when finished, an undertaking which will afford a most valuable addition to the accommodation of the port. This is the cheapest dock in London, the original cost exceeding a little over £5,000 per acre only. But it should be borne in mind that the ground excavated being marsh land, presented a natural nucleus for a dock; and to this fact must be attributed the inexpensive character of this undertaking.

The collier and coasting trade generally has always been a great impediment to the navigation of the river; and I am informed on the best authority that the object of the promoters of the Millwall Dock was to attract this class of business. The colliers had, however, fought shy of the Victoria Dock, which was mainly started for their accommodation, and they proved even more chary of the Millwall Dock. This dock, which was opened in 1864, has, however, been successful in attracting to it a large share of the general business of the port, and has proved no mean competitor to the other docks. The shares which not so very long ago might have been bought for something over £20 the £100 share, now command more than four times that amount. The cost of this dock was about £7,000 per acre.

The next addition to the dock accommodation of the port was the conversion of the City Canal into the existing South-West India Dock, and this view represents the opening of the dock by the entrance of the 'Lufra.' This dock was commenced in the year 1866, and opened on March 5, 1870. Having regard to the fact that when this dock was opened it was little more than a great pond, the warehouses and mechanical appliances being of most limited extent, it is almost impossible to realise the change which it has undergone in the interval. But for the evident newness of the buildings, and the modern character of the mechanical appliances, a stranger would scarcely be prepared to believe that in the vast range of sheds and warehouse accommodation extending all round the dock, he beheld the work of less than five years.

This is another view of the South Dock, seen from the western end, and its crowded condition helps still further to carry on the contrast between the recent and present appearance of this noble dock. The fact is the Company were scarcely prepared for the rush of shipping which immediately followed the opening of this dock. The result has been, and is still, to some extent, a neck-or-nothing race between the urgent demands of the public for more accommodation, and the efforts of the Company to supply it. When it is remembered that many vessels frequent this dock upwards of 350 feet in length, discharging 2,000 or 3,000 tons of cargo to the orders for delivery overside to thirty or forty different consignees, in less time than it would formerly have taken to discharge a hundred-ton billyboy, some idea may be obtained of the efforts which have been put forth by the Company, in so

short a time, to secure the high efficiency necessary to perform work under these conditions. Fancy what our forefathers would have said to a prediction, that the time would come when it would be possible to discharge 1,200 tons of miscellaneous cargo in eight and a half hours, and that samples and accounts would be in the hands of the merchants in the course of the following morning! Impossible as this would have appeared fifty years ago, it is now of frequent occurrence. And, if there are any practical dock officers here they will bear me out in the assertion that, owing to the bitter competition and the insane demand for despatch, dock officials are about as hard-worked, worried, and harried a class of public servants as are to be found anywhere!

This is a view of the West India Import Dock seen from the eastern end, and gives a very excellent picture of the North Quay Warehouses. In these warehouses and on the quays there has been at one time property of the value of £20,000,000 sterling. This dock is 30 acres in extent, and, next to the Victoria, is the largest dock in London.

This is another view of the same dock, seen from the western end, showing the South Quay, which comprises the Rum and Wood Departments. Upwards of 45,000 casks of rum have been warehoused here at one time.

This is a view of the East India Import Dock. Many of you are probably aware that the Company are enlarging the basin of the dock, and constructing another entrance; there being at the present time but one, always a source of risk; as, should it happen, as at Hartlepool in May last, that a ship got wedged in the lock, thereby stopping the exit or entrance of vessels, the inconvenience to business would be indescribable.

This is a view of that part of the London Dock known as the Crescent. This is an exceedingly interesting dock. The Wine Vaults alone may be described almost as one of the wonders of the modern world. One of them is nearly twelve acres in extent, and if you wish to fancy yourself in the Catacombs, without the trouble of a visit to Rome, you cannot do better than see the Wine Vaults at this or the St. Katharine's Dock. Altogether there is storage accommodation for 80,000 pipes of wine and brandy. In fact as the West India Dock is the great depôt for rum and sugar, so this dock is the great depôt for wine and brandy. The Queen's Pipe is also a most

interesting feature of this establishment. Here all condemned goods are destroyed by fire, which at times receives some odd contributions. On one occasion 45,000 pairs of gloves were consigned to the flames, and on another, 900 Austrian mutton hams. Indeed comparatively few people who cursorily scan the frowning walls of the great dock establishments are aware of the varied information and instruction to be gathered within them.

I have thus finished my notice of the entire dock system of the Port of London, for a complete view of which see Appendix [L](#): and a glance at this plan, which has been specially drawn in connection with this lecture, will convey some impression of its magnitude. It will be found very interesting to compare London as it *is* with what it *might have been* had some of the schemes to which I have called your attention, been carried out. The aggregate water-space of the docks of London, when the extensions in progress shall have been completed, will not be less than 560 acres. It should be borne in mind that the whole of these costly undertakings are the birth of the present century. Within the like period we have seen the commerce of the port multiplied nearly seven-fold; and it is no exaggeration to say, that to this marvellous result, the docks, in conjunction with the development of the warehousing system, which they have so much assisted, and to which I should like to have referred, have contributed more than any other agency. I fear that the public generally are rather inclined to lose sight of this fact, which becomes obvious enough after a few moments' consideration. If the tremendous evils to which I have called attention as existing less than a hundred years ago, when the business of the port was not one-seventh of its present magnitude, were the result of overcrowding, it stands to reason that, in the absence of dock accommodation, the increased business must have gone to the out-ports, and, failing the necessary accommodation there, have located itself on the Continent. This, it is unnecessary to say, would have been nothing short of a national calamity. Look at Liverpool, a little fishing village at a time when London absorbed four-fifths of the business of England and Wales. But the dangers incident to the navigation of the Mersey early called attention to the necessity of artificial water accommodation for its shipping. This the people of Liverpool were wise enough to recognise; and who can doubt that Liverpool, which, considering its importance three hundred years ago, has grown even more rapidly than London, owes its marvellous prosperity to its magnificent dock system. Look, again, at Bristol. Time was when Bristol,

as a port, ranked next to London; and why? Simply because of her unrivalled geographical situation and the early establishment of the dock system. Why does Bristol no longer occupy that prominent position? Because, as is generally the case where Nature has done so much, Art, her handmaid, has been less assiduous in her attentions than at London, Liverpool, Hull, and other places; and so, in the race for commercial prosperity, Bristol has been left behind by her younger, and more enterprising, sisters. Clearly Nature intended Bristol to be, what she once was, the most flourishing port on the west coast of England, and no mean rival of London. But the Bristolians appear to have forgotten that large ships need plenty of water, and that the Avon, like some people of unsettled habits, has a strange fancy at times for leaving its bed unoccupied. Could the good people of Bristol be induced to dockise that very erratic stream within a respectable distance of its mouth, I am inclined to believe that many a shipowner who now sends his vessels to London would think it worth his while to escape the dangerous Channel passage by ordering them into Bristol, more especially with an equalization of railway rates. I never visit this grand old city, so rich in historic memories of maritime enterprise, the birthplace of Sebastian Cabot, the real founder of the great Russia Company, the moving spirit amongst the 'Merchant Adventurers,' and 'Grand Pilot of England,' without feeling that, sooner or later, she will again assert that supremacy in the West which is her natural heritage. The new docks at Avonmouth and Portishead are a step in this direction, and cannot fail, if well worked, to command a large share of the business of the Western world. If I am right, then for peaceful, prosperous, contented Bristol there are even greater things in store than she has ever yet dreamed of in her commercial philosophy.

Of course, a sketch of the Thames and its docks would be incomplete without a statement of the Legal Quays, Sufferance Wharves, and Private Warehouses now in existence. Few people are aware of their enormous extent, and it would be difficult to exaggerate the important part which they have played in the history of the port. There cannot be a doubt that the proprietors of these places have availed themselves to the fullest extent of the advantage which they enjoy, under the present state of the law, of being allowed free use of the dock-waters. Undermentioned is a list of them, and I may add that, with very few exceptions, all restrictions as to the goods which may be landed at some of these places have now been removed.

Legal Quays.

Fresh Wharf.	Custom House Quay.
Hammond's.	Brewer's Quay.
Cox's Quay.	Chester's Quay.
Botolph Wharf.	Galley Quay.
Nicholson's Wharf.	

Uptown Warehouses and Vaults.

Aire & Calder.	Metropolitan Warehouse and Vaults.
Beer Lane Vaults.	Mint Street Warehouse.
Bell's Warehouse.	Monastery Warehouse.
Billiter Street Warehouse.	Monument Warehouse and Vaults.
Cooper's Row Warehouse.	Priory Warehouse.
Crown Diamond Warehouse and Vaults.	Red Mead Lane Warehouse and Vaults.
Crutched Friars Warehouse.	St. Dunstan's Warehouse.
Cutler Street and New Street Warehouse.	St. Andrew's Vaults.
Dowgate Hill Vaults.	St. Olave's.
East India Avenue Vaults.	Savage Garden Vaults.
Fenchurch Street Warehouse.	Smith's Warehouse.
George Street Vaults.	Thames Street Vaults.
Globe Yard Warehouse.	Tower Hill Vaults and Warehouse.
Gracechurch Street Vaults.	Trinity Warehouse and Vaults.
Lingham's Warehouse and Vaults.	Water Lane.

Sufferance Wharves, at which certain goods may be landed and warehoused.

Allhallows.	Mark Brown's.
British and Foreign.	Metropolitan.
Brooks'.	New Crane.
Bull.	New Dundee.
Butler's.	Old Swan.
Chamberlain's.	Oliver's.
Commercial.	Paul's Wharf.
Commercial Dock.	Pickle Herring, Lower.
Cotton's and Dépôt.	Platform.
Davis's.	Red Lion and Three Cranes.
Dowgate Dock and Warehouse.	St. Bride's Upper.
Dyers Hall, and Monument Warehouses.	St. Bride's.
Fenning's.	St. John's.
Fishmonger's Hall.	St. Katharine's.
Gun.	St. Olave's.
Gun and Shot and Griffin's.	St. Saviour's.
Hartley's.	Smith's.
Hambro'.	South Eastern.
Hay's.	South Devon.
Hermitage.	Springall's.

Hibernia, New.
Irongate.
London and Continental.
Lucas and Spencer's.

Symon's.
Topping's.
Willson's.

Sufferance Wharves at which certain goods may be landed.

Aberdeen.	Kitchen's, Lower.
Atkin's.	Landell's.
Barnard's.	Mellish's.
Barry & Co.	Metropolitan, New.
Bethell's.	” Upper.
Brandram's, New.	Mill.
Brandram's, Lower.	Miller's.
Broken.	Morton's.
Brook's, Upper.	Newell's.
Brown's.	Noehmer's.
Brunswick.	Orchard.
Burt's.	Ordnance.
Canada.	Patent Fuel.
Carron.	Pearson's.
Chapman's.	Peruvian Guano Works.
Clyde.	Phillip's.
Cole's, Upper.	Plaistow.
” Lower.	Pontifex and Wood.
Coventry.	Pooley's.
Dock Wharf (Regent's Canal Dock).	Powell's.
Dudin's.	Prince Regent's.
Durrand's.	Reed's, Upper.
Eagle.	” Lower.
Fisher's.	Scott's.
Fogg's.	Sharp's.
Foreign Cattle Market.	Stanton's.
Freeman's.	St. George's.
Free Trade, Lower.	Sun.
Garford's.	Sunderland.
Gibb's, Upper.	Surrey Canal Docks.
” Lower.	Tubb's.
Granite.	Union.
Guernsey Granite.	Victoria.
Hall's.	Victoria Wharf.
Harrison's.	Watson's.
Hibernia.	West Kent.
Hudson's.	Whiting's.
King and Queen.	Worcester.

This is a view (see [Frontispiece](#)) of the South-West India Dock seen from the western end. I have had this view taken especially to illustrate the few remarks with which I shall close my lecture. You will observe the accumulation of barges just inside the dock waiting to go out, and you will be able to form some idea of the terrible inconvenience which the presence of these craft must cause in the dock, crowded as it constantly is with shipping; and it so happens that the accumulation of these barges is greatest when there is the least room for them. I must ask you to imagine that behind these barges there are several ships waiting to leave the dock, and that *outside* the dock there is a similar accumulation of lighters and large vessels waiting to come in. The presence of these barges, as shown in this view (which it must be remembered was photographed on the spot, and with no thought of using it in this connection), will help to convey some idea of the delays and risk which they cause in the docking and undocking of vessels—a delay and risk which, as I shall presently explain, the Dock Companies are compelled to incur altogether *free of charge*.

Now you will have observed that all the earlier Dock Companies started with a monopoly for twenty-one years of the shipping from certain parts of the world. You will also have noticed that the object of the Government in granting these monopolies was twofold; first, by compulsorily clearing the river of a large proportion of the shipping to remove the facilities for plunder afforded by the lighterage and unprotected condition of goods; and, secondly, to secure for the docks some equivalent for the money invested in them. Now what do these facts indicate? Clearly that the Government saw the necessity of legislative interference in the interest of the port, and that, to render that interference operative, they must offer an inducement to the public to invest their money. Thus, you will observe that the Dock Companies sprang into existence, not so much independent speculations, as a response to an appeal from the Government of the day on behalf of the port.

But this compulsory alienation of certain shipping to certain docks, of course involved the withdrawal from the Legal Quays of the profits on the warehousing of goods left in the docks, and from the lightermen much of the valuable carrying trade represented by the shipping covered by the monopolies. It also affected a number of minor collateral interests. These were, of course, entitled to compensation, and they obtained it. The total

amount of the claims received was little short of £4,000,000. As is generally the case under the circumstances, many of these claims were grossly exorbitant, and others had no foundation at all. The Government were, however, disposed to take a liberal view of the situation, and, as I have already told you, actually awarded out of the Consolidated Fund not less than £1,681,000, which, of course, included the sums paid for the purchase of the Legal Quays.

But the Government not only made ample *money compensation* for the rights which they invaded by granting the dock monopolies, they also made access to the vessels in the docks as free as if they had still discharged in the river; that is to say, the waters of the dock were to be accessible, free of charge, to anybody who wished to bring in lighters, for the purpose of taking goods or ballast to or from any vessels lying in them. The reason for this is obvious. Virtually the Government said to the Dock Companies: ‘As we have compelled ships to enter your docks, you must not tax certain lighterage to or from those ships.’ And this was perfectly reasonable and fair so long as such compulsion existed; that is to say, so long as the Dock Companies got a *quid pro quo* for the free use of the dock waters by the goods secured to them by the monopolies. But so soon as ships became free to go where they pleased, and the docks were left to compete with the wharves on even terms, this restriction upon their obvious right to charge for the use of their property should have been removed also. It must always be borne in mind that the *basis* of the arrangement in regard to the *free use of the dock waters* was the *compulsory alienation* of certain shipping to the docks, which had hitherto been *free to discharge in the river*; hence it is obvious that so soon as that freedom was restored to the shipping referred to, the *raison d’être* of the restriction upon the docks ceased to exist. Mr. Lindsay, an impartial witness, from whose valuable work I have already ventured to quote, cannot help noticing this anomaly. He remarks: ‘These privileges, granted originally to stifle opposition, they (the lightermen) still retain to their gain, and that of the wharfingers, but to the loss of the Companies. Surely, when the monopoly of the Companies had expired—a monopoly to which, for the time, they were fully entitled, considering the service they had rendered to the Crown, in the protection of the revenue—these privileges to the barge-owners should also have been withdrawn.’

Now the liabilities of the Dock Companies of London represent a total of upwards of £15,000,000. More than one-half of this has been spent in providing the water accommodation of the several docks, and the interest on it of course represents a vast permanent tax upon the revenues of the Dock Companies, towards which, although enjoying full use of the docks, the wharves do not contribute anything. The dock waters are supposed to be paid for by the dues on shipping. As a matter of fact this is not the case, although if it were, the claim of the Dock Companies to be paid for services altogether distinct from the operation of docking and undocking ships, would be in no respect weakened. For a considerable time more than half the goods brought into the docks by ships have been taken out of them again for warehousing elsewhere, and upon the tens of thousands of lighters employed in this service constantly inside, entering, and leaving the docks, the Companies are prohibited from charging one penny; so that, coupling this circumstance with the fact that the dues of the Company are levied upon the register and not upon the gross tonnage of vessels, it will be seen that, practically, the income from the shipping rates is reduced to considerably less than one-half the amount which it nominally represents.

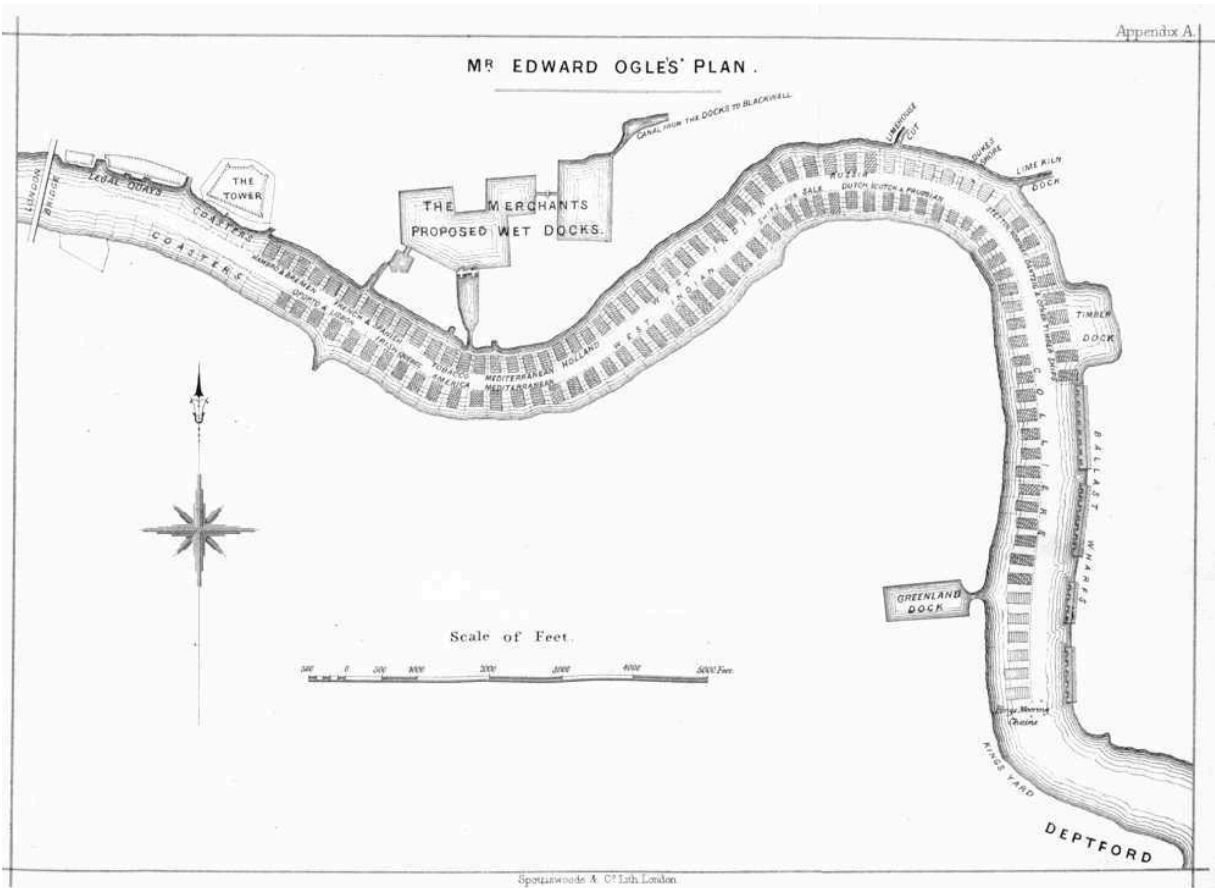
But this is not the only hardship experienced by the Companies. The expense and inconvenience in working the docks owing to the number of these barges in them cannot be measured by money loss. In the West India Docks alone there have been 500 barges at one time occupying the water space when most urgently needed for the working of ships. The delays caused in the docking and undocking of vessels in consequence of the glut of craft waiting at tide time to enter or leave the docks are sometimes attended with considerable danger. In fact, this view represents the normal condition of the dock entrances whenever there is an influx of shipping. Every barge entering and leaving the docks, of course entails a distinct service on the Companies for their own protection, for which the parties so compulsorily served pay nothing; and the most startling feature in the case is the fact of the growing magnitude of this burden. Some idea of its extent may be seen when I state that the number of barges which entered the East and West India Docks alone in 1869 was 51,985. In 1874 the numbers had rushed up to 61,390—an increase of 20 per cent. in five years. Surely these facts speak for themselves, and bear me out in the assertion that there are no other public bodies in London which, having rendered such incalculable service to the port, have been left in such an anomalous position. In fact,

you will see that the Dock Companies are pretty much in the position of a man who, having been induced to build a house on a particular spot on the understanding that a tenant would always be found for it, not only loses his tenant but finds himself compelled to maintain his house in a high state of efficiency, and to keep a large staff of servants to wait upon anybody who may take a fancy to anything in it, without so much as feeing the porters.

In dealing with this question you will observe that I have carefully abstained from any arguments as such, preferring to leave the facts to speak for themselves. I might have told you that before the docks were in existence it was the practice at the wharves to charge half-wharfage rates on goods passing over them; and I might have argued that if this were recognised as right in a free river, surely the Dock Companies should not be refused some equivalent return for the use of the dock waters—a private property which has cost many millions. From this argument and many kindred arguments I have abstained; but in closing my lecture I must be permitted to anticipate just one objection. It may be said, why did not the Dock Companies move in this matter when their monopolies expired? My answer will be appreciated by all practical men. When the dock monopolies expired, the business of the port was not one-fourth of its present magnitude, to say nothing of the fact that the lighterage to the docks was so insignificant that it entailed no practical inconvenience. The force of this will be seen when it is borne in mind that the tonnage of inward shipping, British and foreign (exclusive of the coasting trade), which entered the port in 1827—the date of the expiration of the monopolies—was only 990,170 tons; and as by far the greater portion of the cargoes brought by vessels which entered the docks was left in them for warehousing, the lighterage was, necessarily, of very limited extent, and remained so, until the repeal of the monopolies began to take practical effect in the diversion of goods from the docks. But, side by side with the gigantic increase in the commerce of the port, there has been a corresponding increase in this gratuitous lighterage business. Its magnitude is seen at a glance when I say that in the year 1874 the tonnage of British and foreign inward shipping (exclusive of the coasting trade) had reached the enormous total of 4,671,676 tons. Of this vast tonnage nearly 3,500,000 tons discharged in the docks, and of this 3,500,000 tons of register tonnage, probably representing 4,000,000 tons gross of goods, more than half was removed from the docks. Thus, it will be seen that to argue that because the Dock Companies, *when their monopolies*

expired, did not regard the free use of their waters as a grievance grave enough to call for the interference of Parliament, ought not to complain and have no right to such interference *now*, is to ignore the fact, patent to everybody, that circumstances have so entirely changed the relations which formerly existed between docks and wharves that no analogy exists to warrant such a conclusion. It was utterly impossible for the Dock Companies, fifty years ago, to foresee this gigantic growth in the business of the port, and the complete revolution which has since taken place in the imperial tariff, by which the docks have been so prejudicially affected. There cannot be a doubt that in continuing this concession without the protection of a monopoly, the Dock Companies made a great mistake; but it is one which a grasping and selfish policy should not be suffered to perpetuate. Both competitors for the trade of the port, the docks should be as free from unfair restrictions as the wharves; and it is to be hoped that before long the love of fair play, and hatred of oppression, of which we Englishmen claim to be so proud, will find expression in a great practical remonstrance, in which this burdensome tax upon the vitality and prosperity of the dock establishments, to the foundation of which London mainly owes its marvellous development as the great entrepôt for the commerce of the world, will be indignantly swept away.

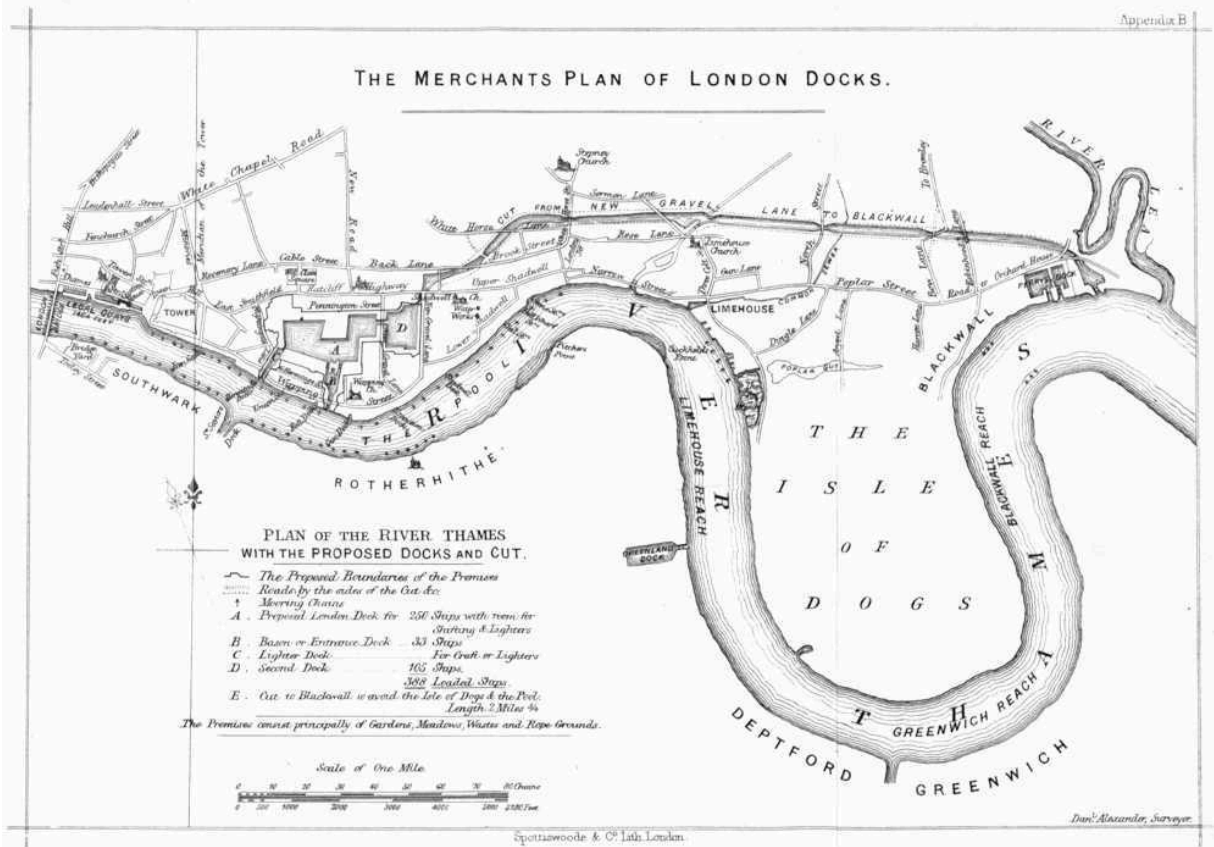
I will detain you no longer. What I have stated is but a sketch of a portion of a subject which a dozen lectures would not exhaust; and, as I only put it forward as a sketch, I must ask you, in making your criticism, to bear that fact in view. I have simply sought to lay before you a few facts not generally known and not easily accessible, and which I hope will not fail to invest the river and the docks with additional interest to all whose business or pleasure may be affected by either. I can only trust that you have found them interesting, and that you may perhaps find them useful.



Appendix A.

MR. EDWARD OGLES' PLAN.

Spottiswoode & Co. Lith. London.



Appendix B.

THE MERCHANTS PLAN OF LONDON DOCKS.

PLAN OF THE RIVER THAMES WITH THE PROPOSED DOCKS AND CUT.

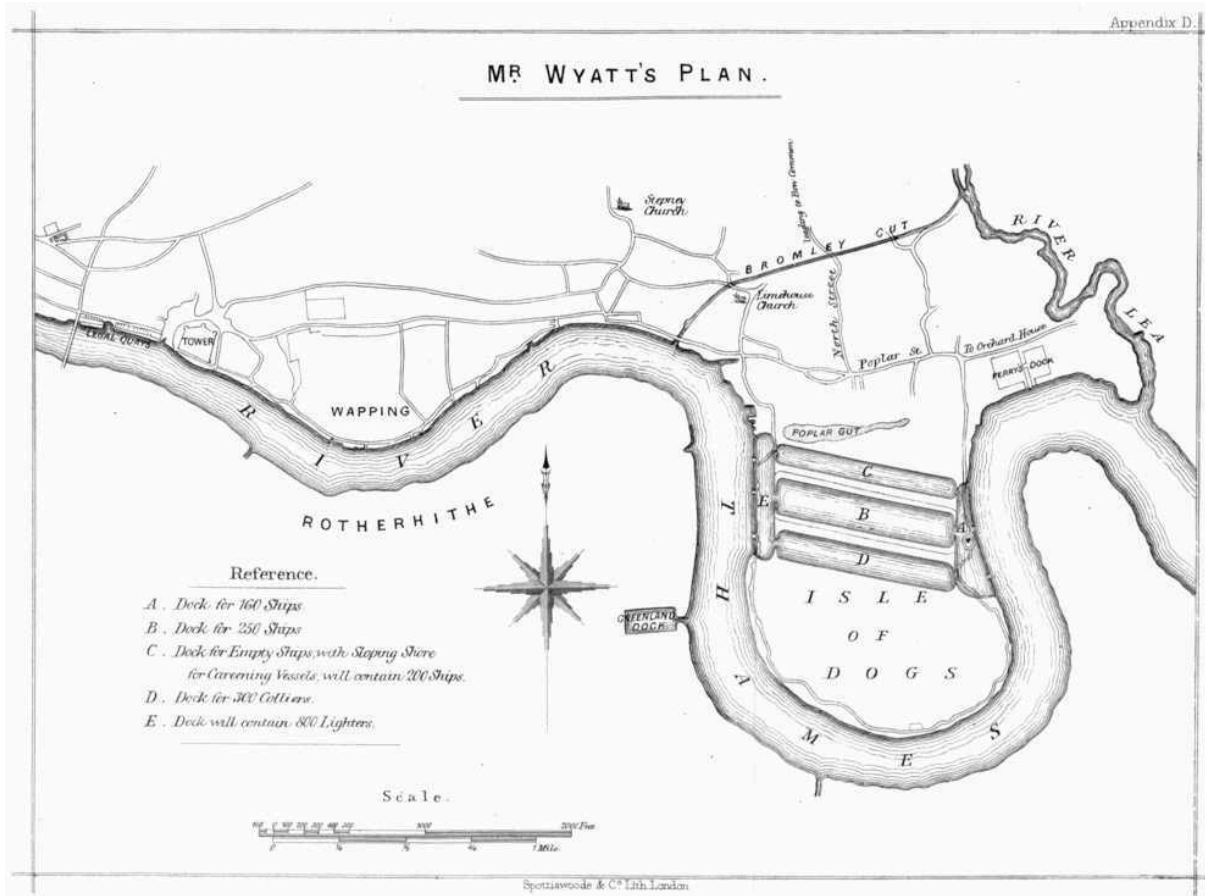
Dan'l Alexander, Surveyor.

Spottiswoode & Co. Lith. London.

Appendix C.

CORPORATION SCHEME.

Spottiswoode & Co. Lith. London.

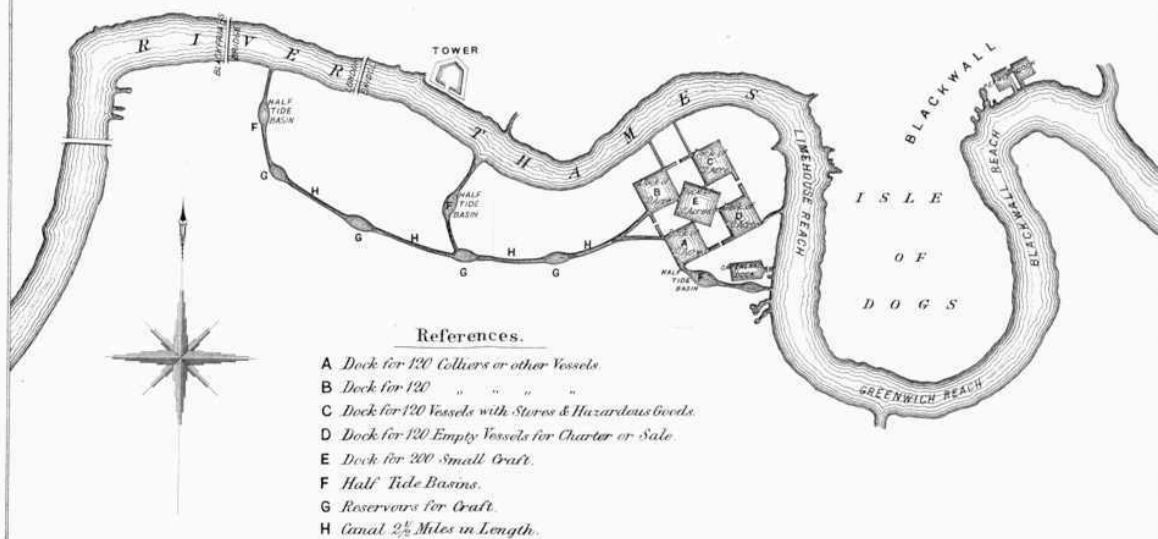


Appendix D.

MR. WYATT'S PLAN.

Spottiswoode & Co. Lith. London.

SOUTHWARK SCHEME.

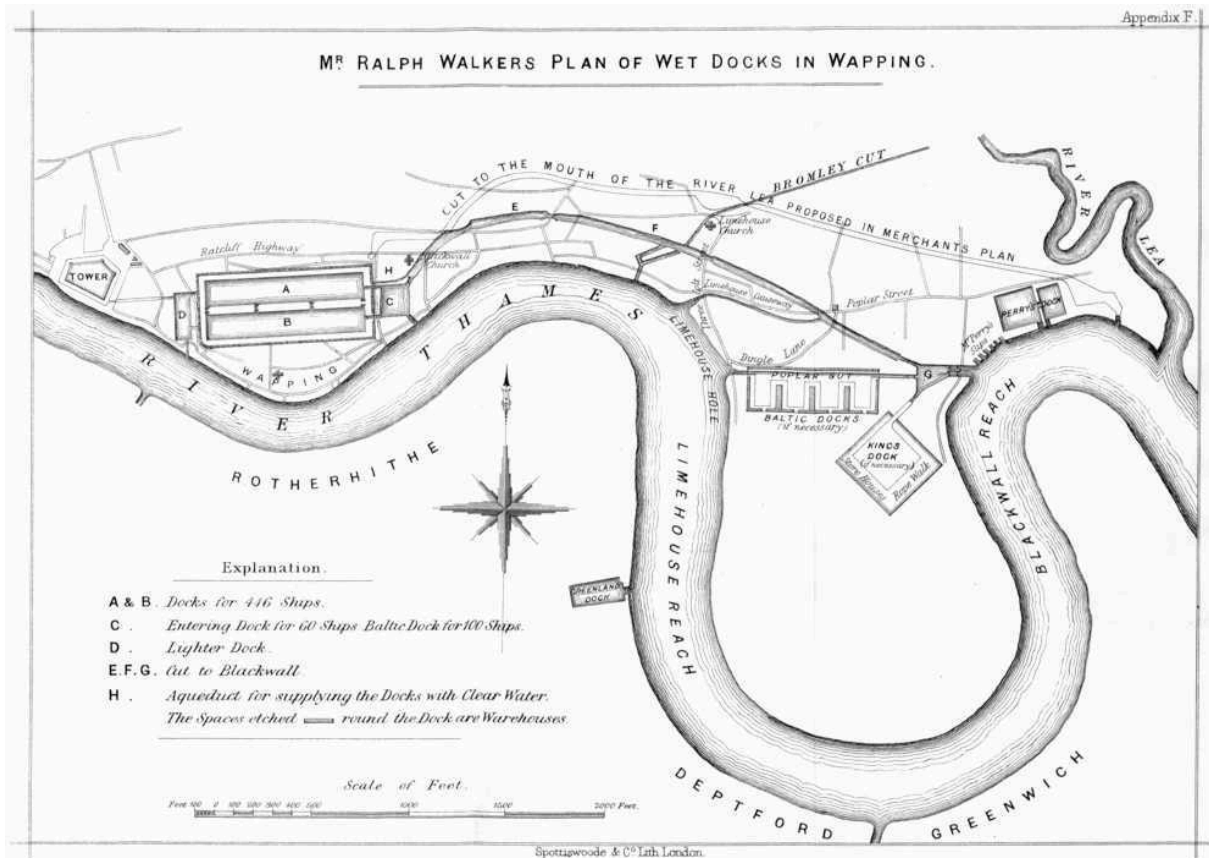


Spottiswoode & Co. Lith. London.

Appendix E.

SOUTHWARK SCHEME.

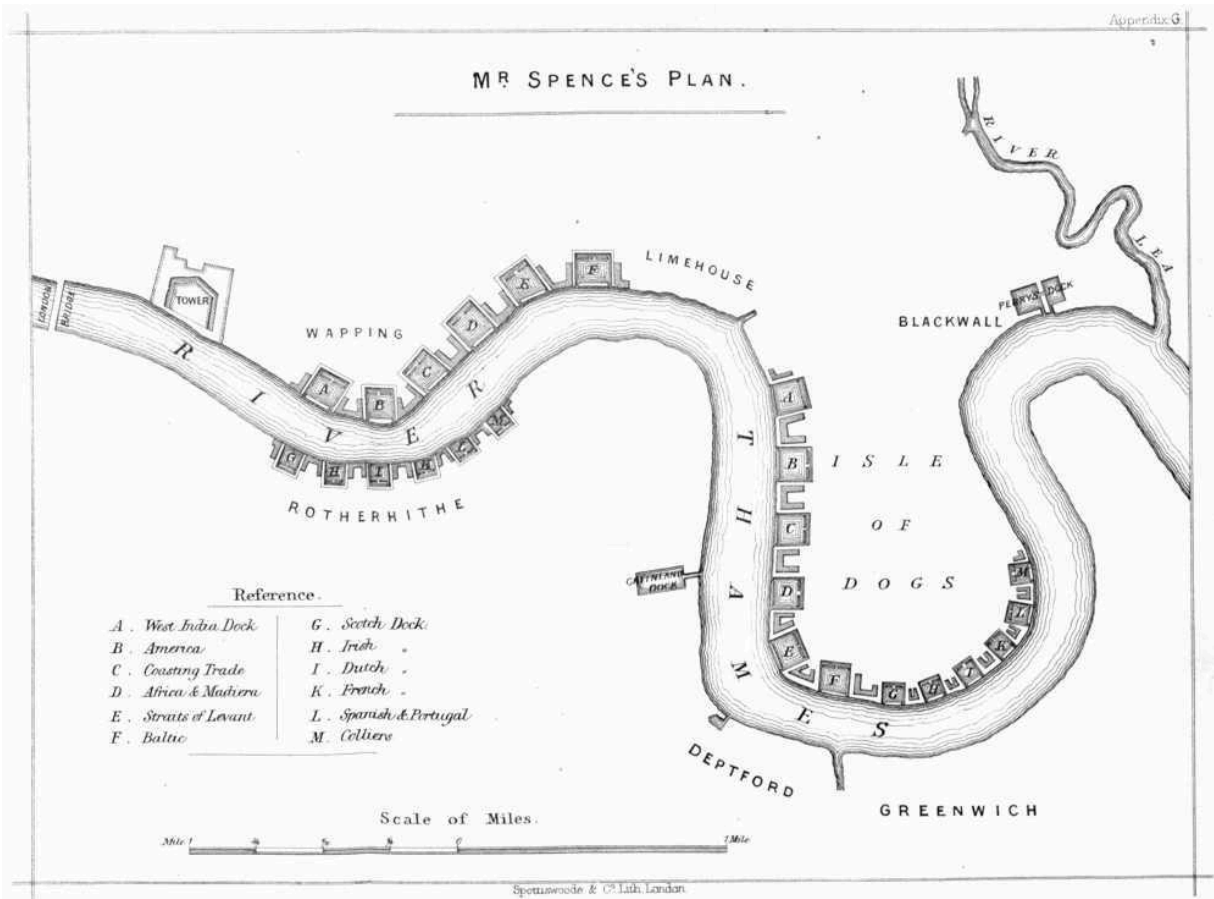
Spottiswoode & Co. Lith. London.



Appendix F.

MR. RALPH WALKERS PLAN OF WET DOCKS IN WAPPING.

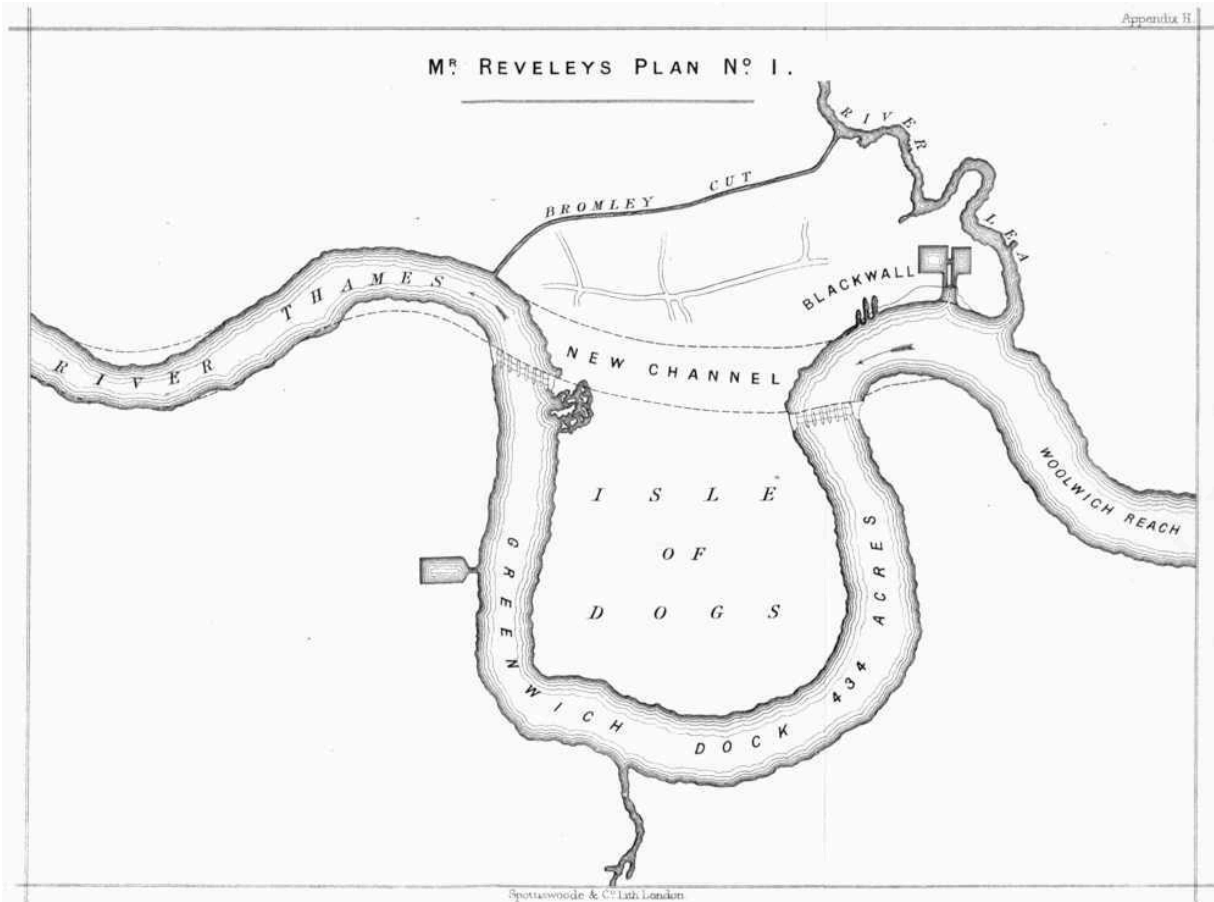
Spottiswoode & Co. Lith. London.



Appendix G.

MR. SPENCE'S PLAN.

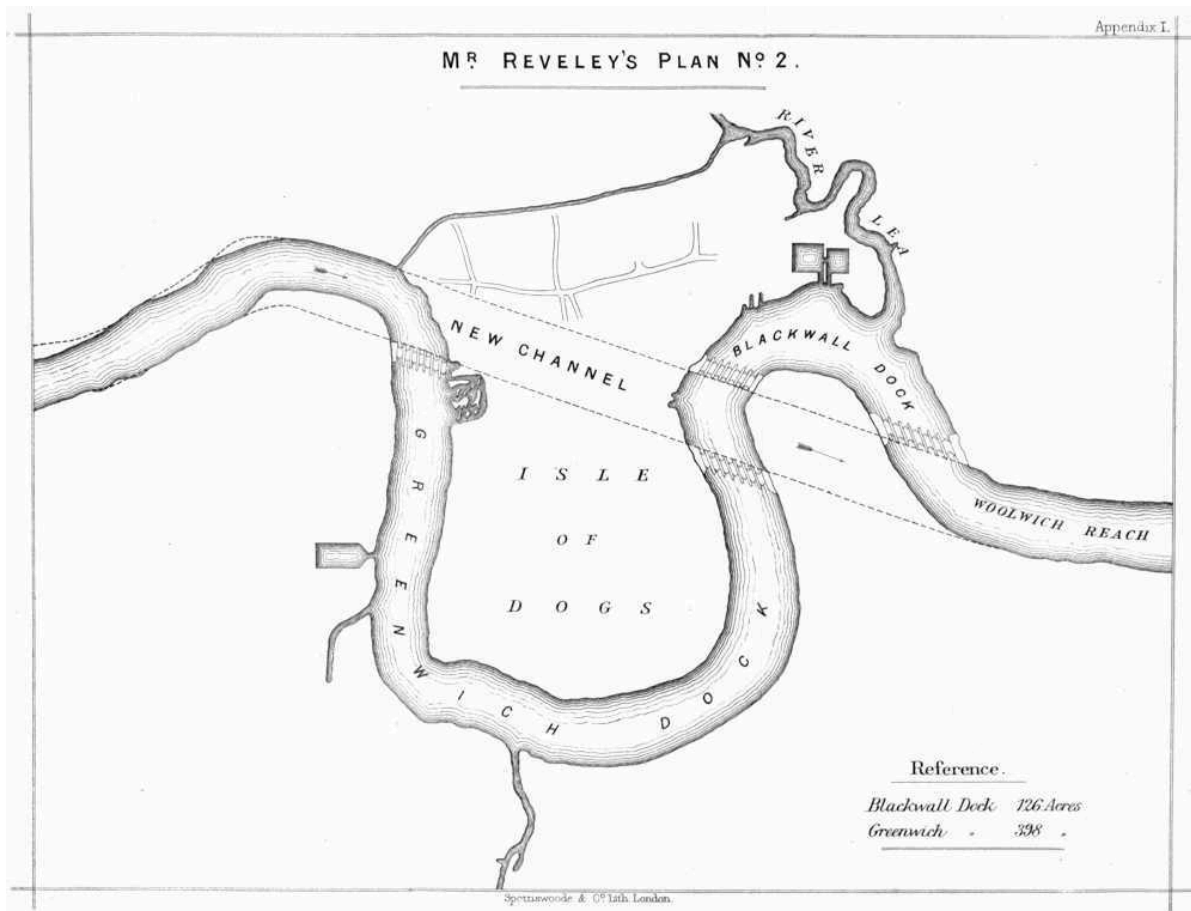
Spottiswoode & Co. Lith. London.



Appendix H.

MR. REVELEYS PLAN NO. 1.

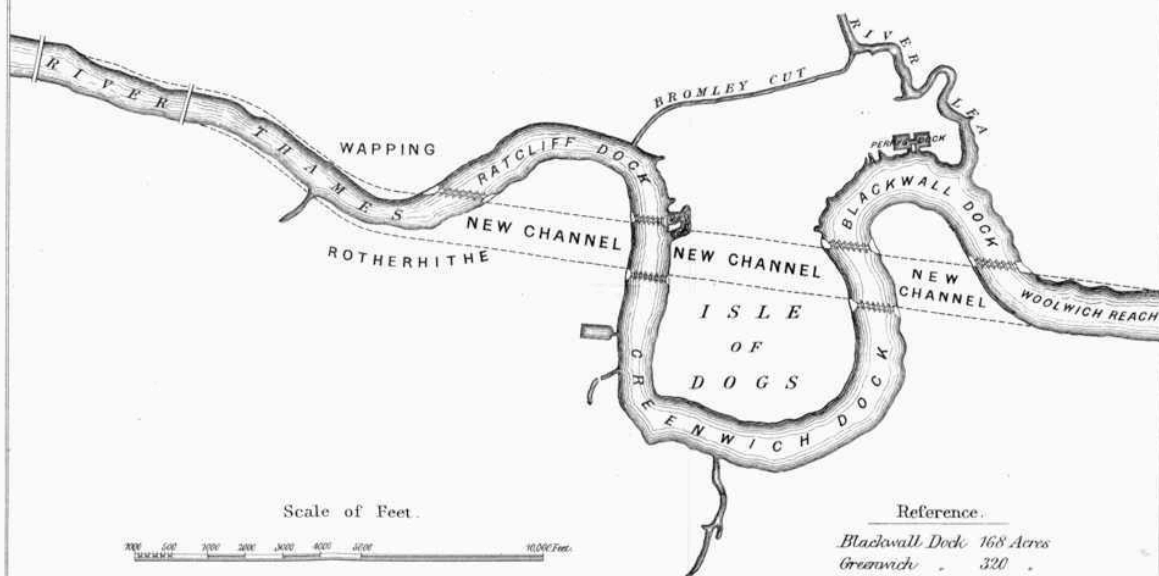
Spottiswoode & Co. Lith. London.



Appendix I.

MR. REVELEY'S PLAN NO. 2.

Spottiswoode & Co. Lith. London.

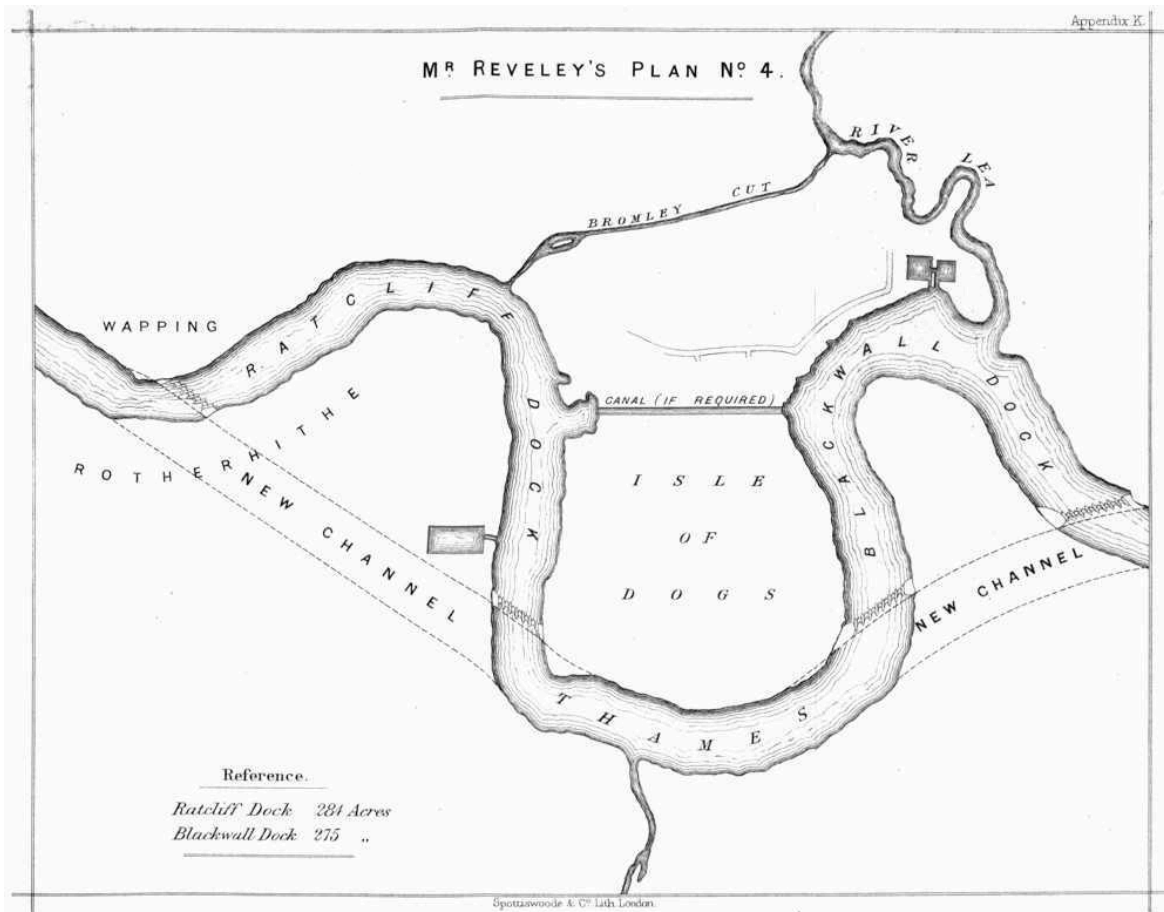
MR. REVELEY'S PLAN N^o 3.

Spottiswoode & Co. Lith. London.

Appendix J.

MR. REVELEY'S PLAN NO. 3.

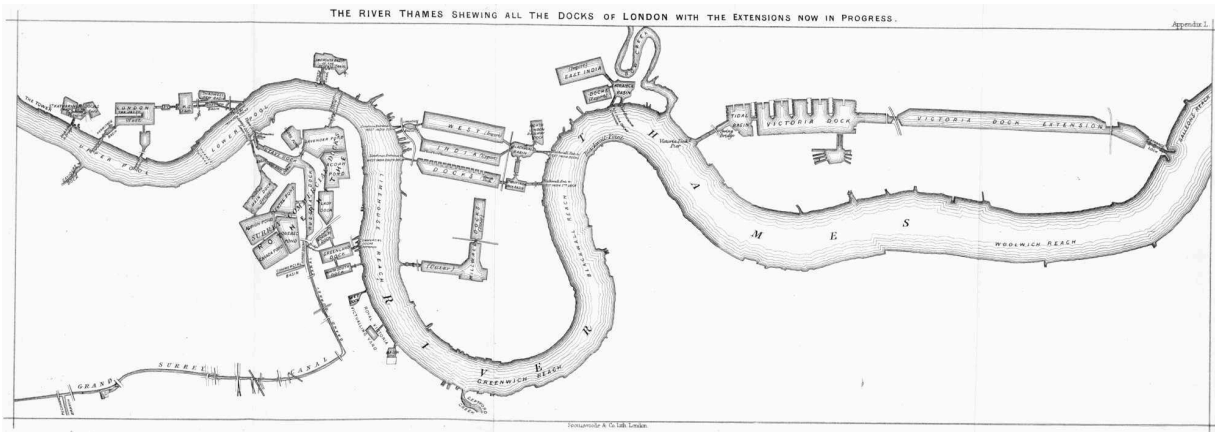
Spottiswoode & Co. Lith. London.



Appendix K.

MR. REVELEY'S PLAN NO. 4.

Spottiswoode & Co. Lith. London.



Appendix L.

THE RIVER THAMES SHEWING ALL THE DOCKS OF LONDON WITH THE EXTENSIONS
NOW IN PROGRESS.

Spottiswoode & Co. Lith. London.

*** END OF THE PROJECT GUTENBERG EBOOK THE THAMES
AND ITS DOCKS ***

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